



RS6x_7x RadarAnalysisTool User Guide

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Revision History

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1.3	Added a description of the FMCW frame structure; Renamed micro-motion point cloud to presence point cloud; Added supplementary notes for excessively small range configurations; Added supplementary notes about the minimum Frame period; Added supplementary descriptions of DOP FFT EN and RANGE FFT EN.	2026-6-27
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1 Overview

This document describes how to use RadarAnalysisTool with RS6x_7x-series mmWave radar sensor devices and provides answers to frequently asked questions.

This document applies to demo presentation and engineering-mode debugging for RS6x_7x-series mmWave radar sensor devices.

2 Instructions

2.1 Equipment Preparation

This section describes the hardware and software setup and preliminary preparations. The firmware [used by RadarAnalysisTool is located at /Software Kit/04 Image/radar_analysis.img.](#)

2.1.1 Firmware Burn (Optional)

Note: If Firmware Burn has already been completed for the Device, skip this step and proceed directly to Section 2.1.2, Device Connection.

1、Hardware setup:

Hardware checklist:

- (1) USB serial cable (Type-C)
- (2) Development board (baseboard and module daughterboard)

Hardware setup procedure:

- (1) With power disconnected, insert the module daughterboard into the baseboard slot. Observe the following precautions:
 - Ensure that the pins are correctly aligned.
 - For the MRS6130_P1806, observe the insertion orientation shown in Figure 2-2.
 - Do not install the MRS6130_P1812 and MRS6130_P1806 modules at the same time.
- (2) Install jumper caps on VCC Supply and V13 Supply. Set the one-position Mode Select DIP switch to OFF. Set positions 1 through 3 of the UART0/UART1/SPI/IIC Select DIP switch to ON. Set VDDIO Supply to VDD3V3 and 5V Supply to ON.
- (3) Supply power to USB COM through USB. The orange power LED illuminates.

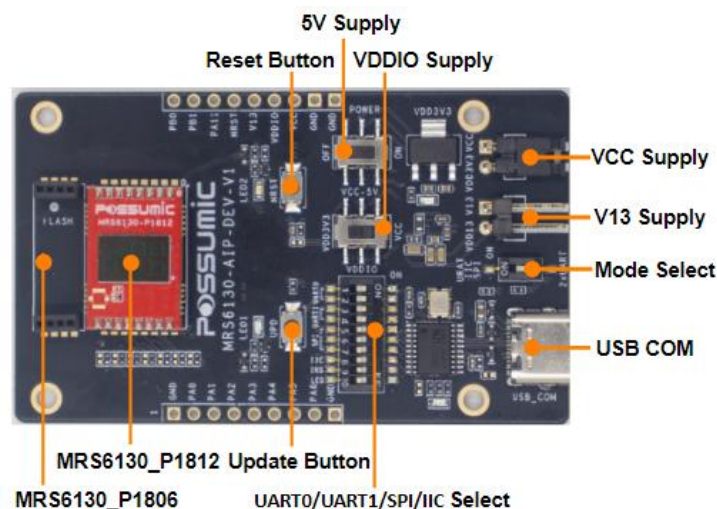


Figure 2-1 Hardware Connection Between the MRS6130-P1812 Module and Development Board

Note:

- (1) When using the MRS6130_P1812 module, do not install a jumper cap on V13 Supply.
- (2) Mode Select configures the USB bus bridge. When set to ON, the hardware interfaces are UART0/SPI/IIC. When set to OFF, the hardware interfaces are UART0/UART1.
- (3) Install the CH343SER driver. If Mode Select is set to OFF, Windows Device Manager displays the serial ports as USB Serial Device (COMx1) and USB Serial Device (COMx2). If Mode Select is set to ON, only UART0 is displayed as USB Serial Device (COMx1).

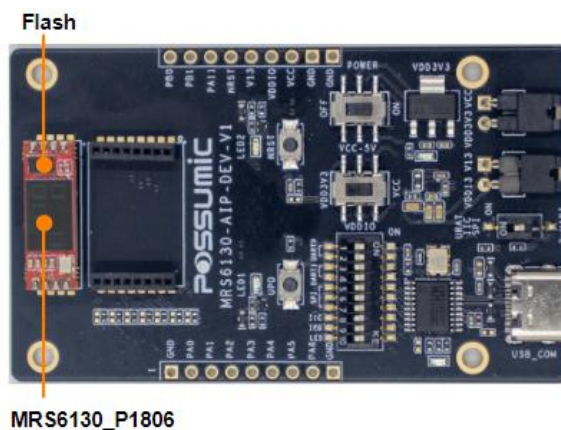


Figure 2-2 Hardware Connection Between the MRS6130-P1806 Module and Development Board

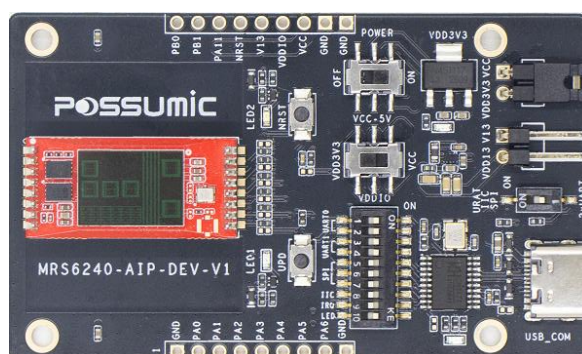


Figure 2-3 Hardware Connection Between the MRS6240-P2512 Module and Development Board

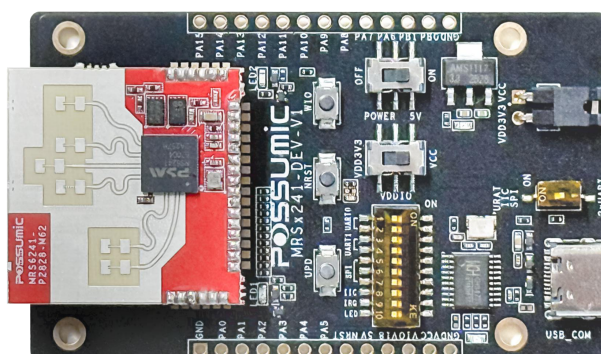


Figure 2-4 Hardware Connection Between the MRS6241-P2828_M62 Module and Development Board

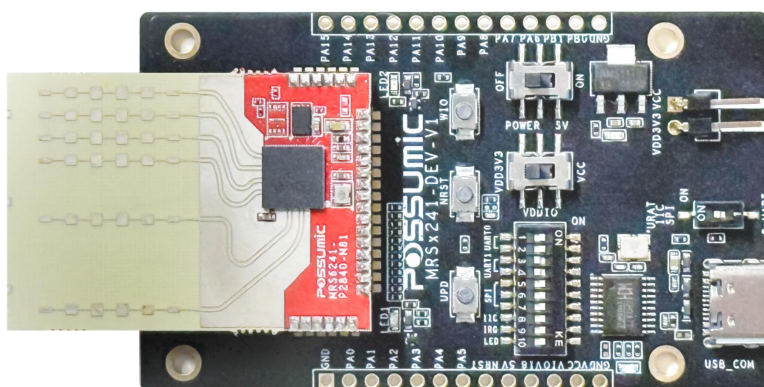


Figure 2-5 Hardware Connection Between the MRS6241-P2840_M81 Module and Development Board

Note: The preceding hardware connections enable two UART channels (UART0 and UART1). RadarAnalysisTool also supports UART_SPI communication. To use UART_SPI, set the one-position Mode Select DIP switch described in hardware setup step 2 to ON, set positions 1, 2, 5, 6, 7, and 8 of the

UART0/UART1/SPI/IIC Select DIP switch to ON, and leave all other settings unchanged. The DIP-switch settings for both modes are listed in the following table.

Table 2-1 DIP-Switch Settings for Hardware Interface Modes

	UART0/UART1/SPI/IIC Select										Mode Select
Switch Position	1	2	3	4	5	6	7	8	9	10	1
UART0 UART1	ON	ON	ON	OFF	OFF	OFF	OFF	OFF	OFF	-	OFF
UART SPI	ON	ON	OFF	OFF	ON	ON	ON	OFF	ON	-	ON

In the two modes listed above, SPI supports transfer rates up to 60 Mbps, subject to CPU frequency, and provides optimized data transfer. UART supports up to 2 Mbps. For greater transfer bandwidth, set the development board to SPI mode. Note that the MRS6130-P1806 module does not support SPI and must use dual UART.

2. Select the hardware platform model.
3. Select the control serial port for the hardware platform. For dual UART, select the USB Serial Device (COMx) port with the larger COM number. For UART_SPI, only one USB Serial Device (COMx) port is available; select that port.
4. Click Select Firmware and select the firmware file.
5. Ensure that the development board is powered off.
6. Press and hold the UPD button on the development board.
7. Power on the development board. After at least 2 seconds, release UPD. The module enters burn mode.
8. Click Burn. RadarAnalysisTool automatically connects to the Device and burns the firmware; you do not need to click Connect Device manually.
9. A successful result is shown in Figure 2-6. After the burn is complete, power-cycle the development board to activate the new firmware.

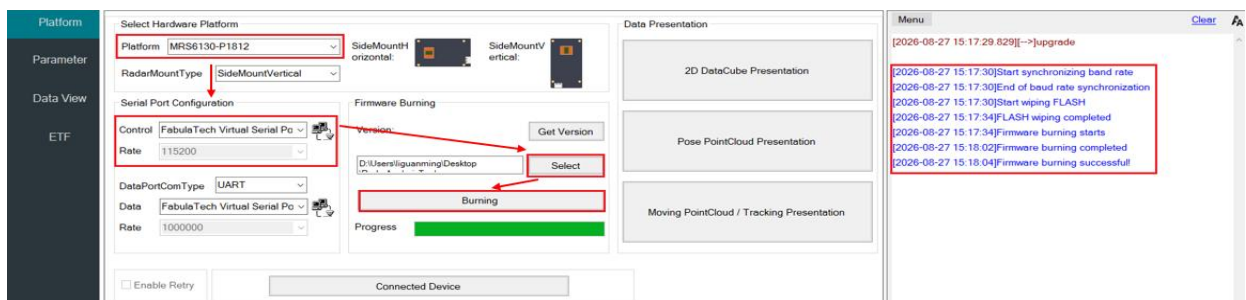


Figure 2-6 Firmware Burn Page

2.1.2 Device Connection

The Platform Configuration pages are shown in Figure 2-7 and Figure 2-9. Connect the Device as follows:

1. Select the hardware platform model and the radar mounting orientation. Select the mounting orientation according to the physical orientation of the development board.
2. Select the control serial port for the hardware platform. This port communicates with the radar module:
 - (1) For dual UART, select the USB Serial Device (COMx) port with the larger COM number.
 - (2) For UART_SPI, only one USB Serial Device (COMx) port is available.
3. Select the data-port communication interface for the hardware platform. This port transfers data reported by the radar module:

- (1) For dual UART, select the USB Serial Device (COMx) port with the smaller COM number. RadarAnalysisTool sends the data-port baud-rate setting to the radar and communicates at that rate. For serial-port compatibility, use 1 Mbaud whenever possible.
 - (2) For UART_SPI, select the port whose name contains CH347T.M1:USB2.0ToSPI&IIC.
- 4、Click Connect Device. The connection result appears in the information pane on the right.
 - 5、After the Device is connected, it automatically switches the data-port baud rate. A successful setting is indicated in the LOG text box, as shown in Figure 2-8 and Figure 2-10.

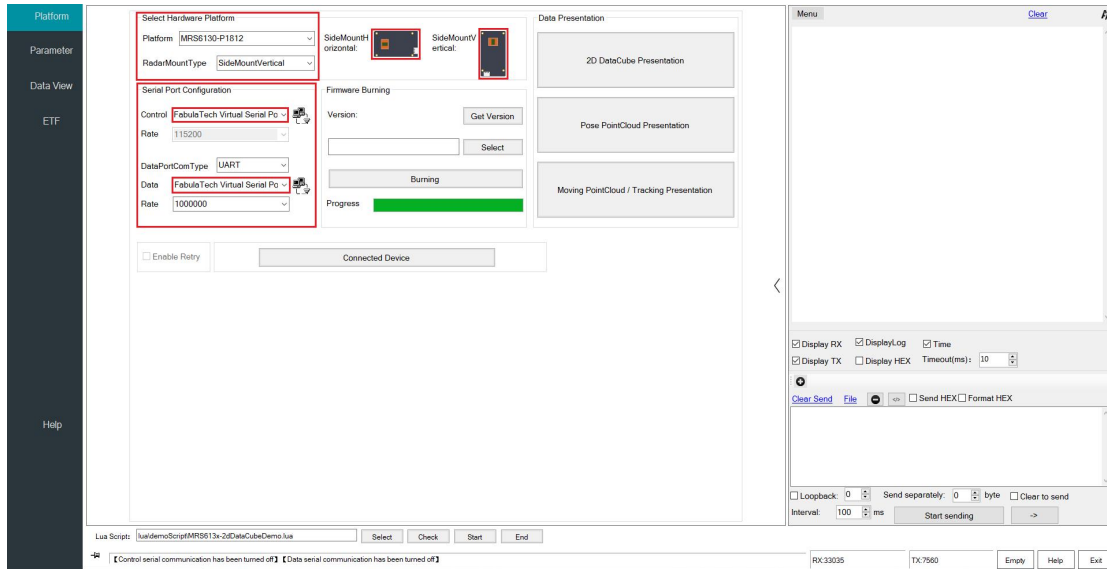


Figure 2-7 Device Connection Page (dual-UART)

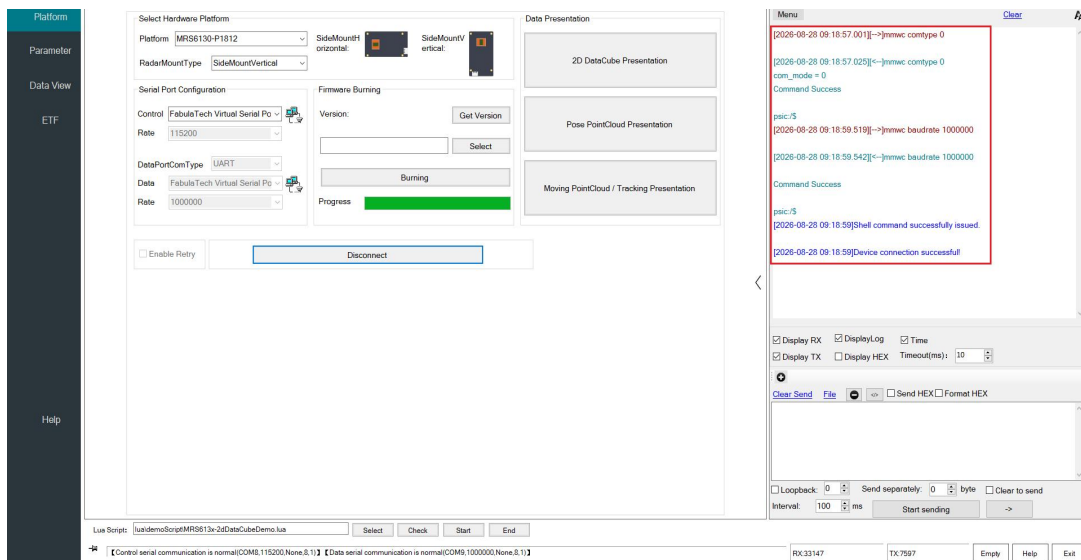


Figure 2-8 Successful Device Connection (dual-UART)

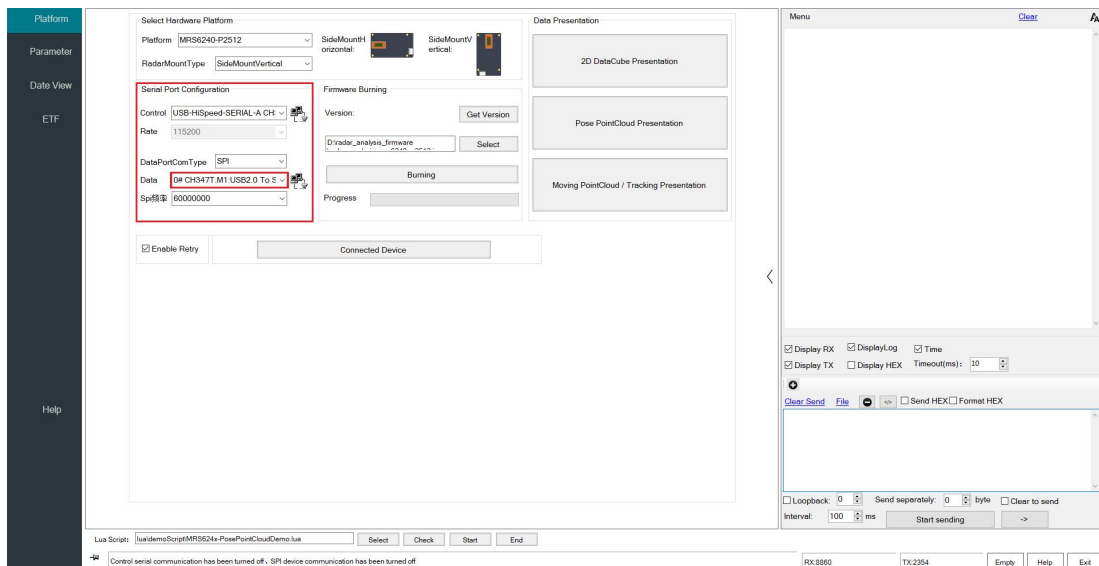


Figure 2-9 Device Connection Page (UART_SPI)

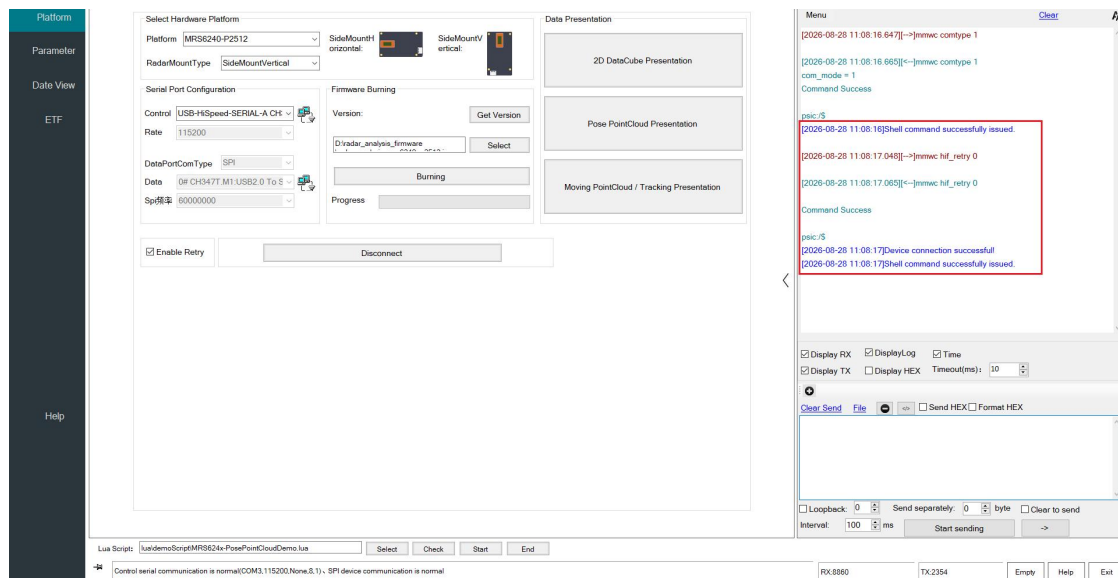


Figure 2-10 Successful Device Connection (UART_SPI)

2.2 One-Click Start

This section describes the 2D DataCube Display, Presence Point Cloud Display, and Motion Point Cloud/Tracking Display functions.

2.2.1 2D DataCube Display

After connecting to the Device as described in Section 2.1, perform the following steps:

1. Click 2D DataCube Display to start the range-spectrum and Doppler-spectrum displays, as shown in Figure 2-11.
2. The software automatically switches to Data View. Select the view to display: Range Spectrum, Doppler Spectrum, Range-Doppler Heatmap, or IQ Plot. Under Display Layout, you can display multiple plots simultaneously, as shown in Figure 2-12. For details, see Data View Display (DataCube).
3. This display uses a fixed one-click radar configuration that cannot be modified. The configuration is shown in Figure 2-13.
4. Note: To perform another operation or change the configuration while the radar is running, first click Stop at the bottom of the Parameter Configuration or Data View page.

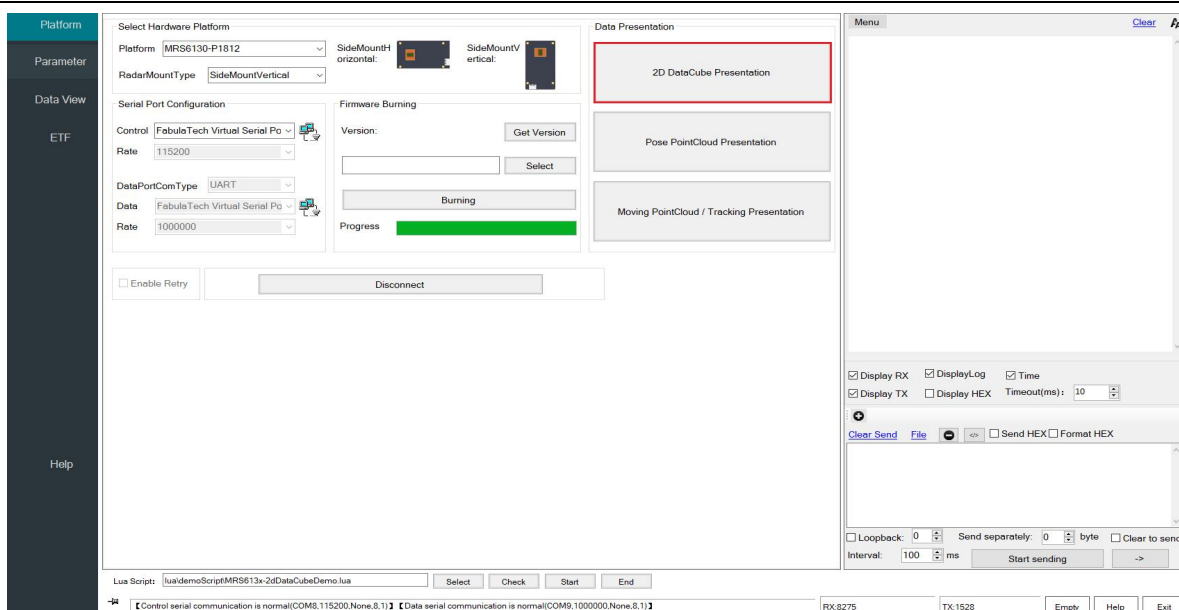


Figure 2-11 2D DataCube Display

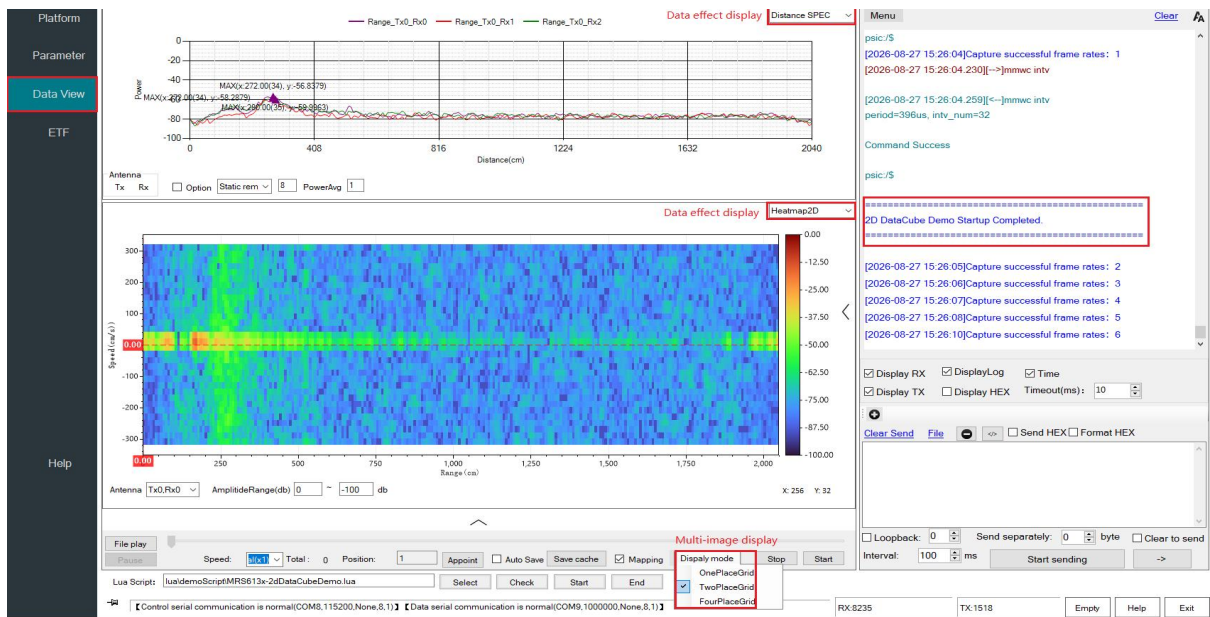


Figure 2-12 Data View

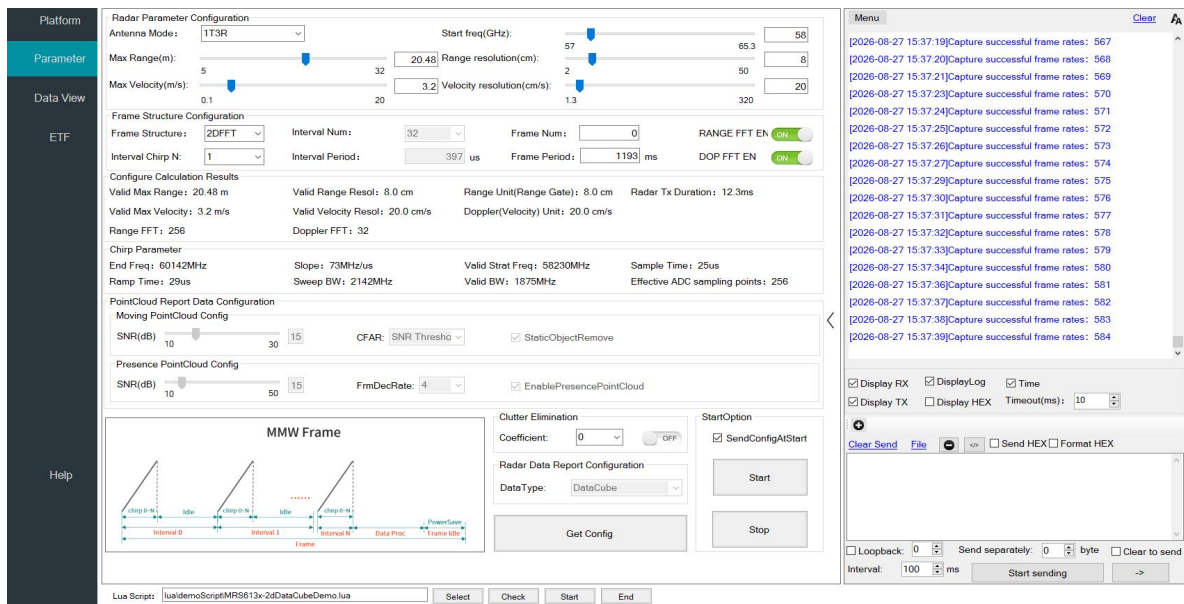


Figure 2-13 2D DataCube Default Configuration Page

2.2.2 Presence Point Cloud Display

After connecting to the Device as described in Section 2.1, perform the following steps:

1. Click Presence Point Cloud Display to start the presence-point-cloud demo.
2. The software automatically switches to Data View and displays the presence point cloud in orange by default, as shown in Figure 2-14. For details, see Data View Display (Point Cloud).
3. This display uses a fixed one-click radar configuration that cannot be modified. The configuration is shown in Figure 2-16.
4. Note: To perform another operation or change the configuration while the radar is running, first click Stop at the bottom of the Parameter Configuration or Data View page.

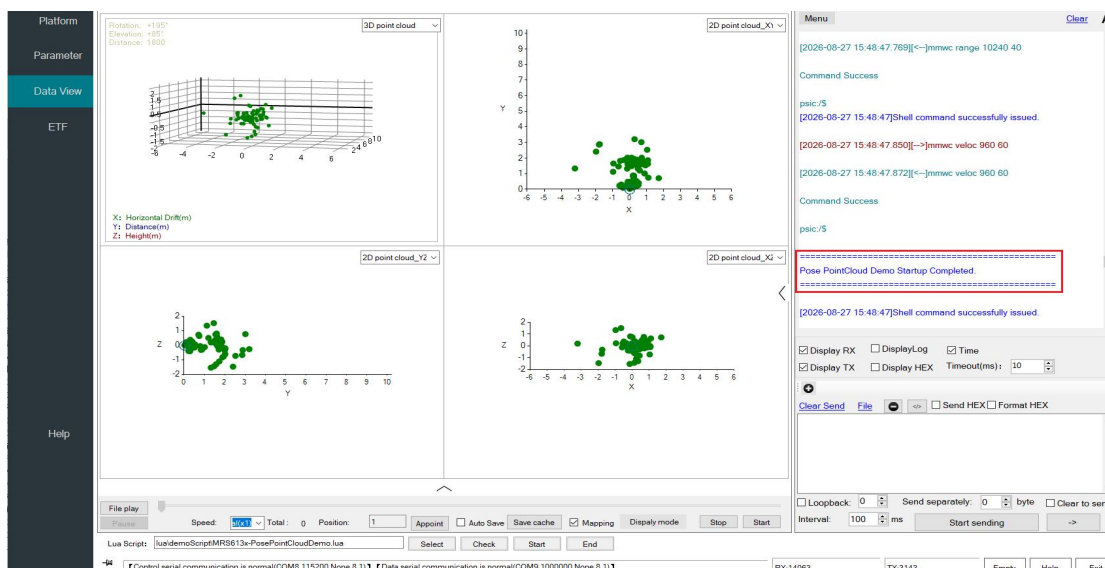


Figure 2-14 Presence Point Cloud Data View

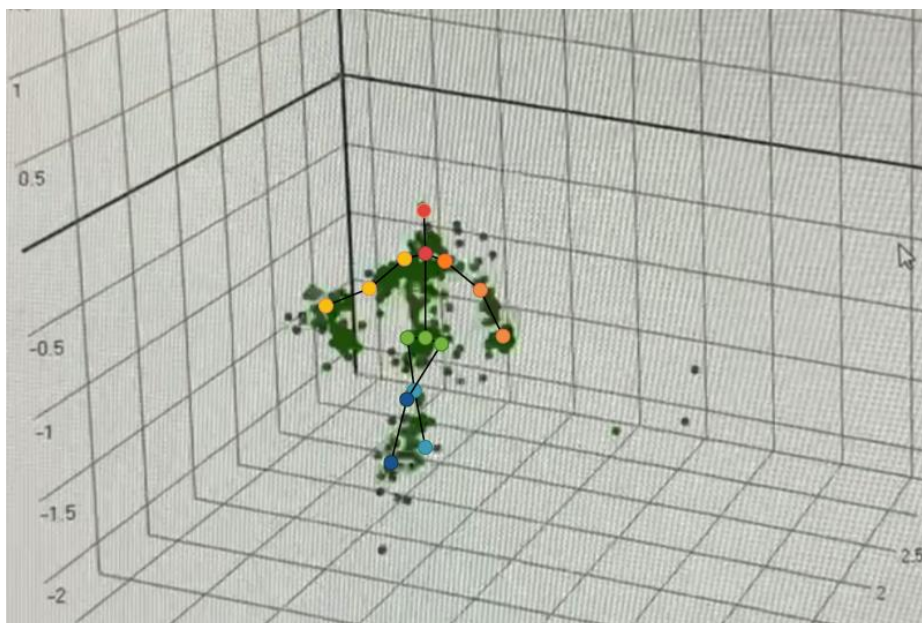


Figure 2-15 Presence Point Cloud Typical Results

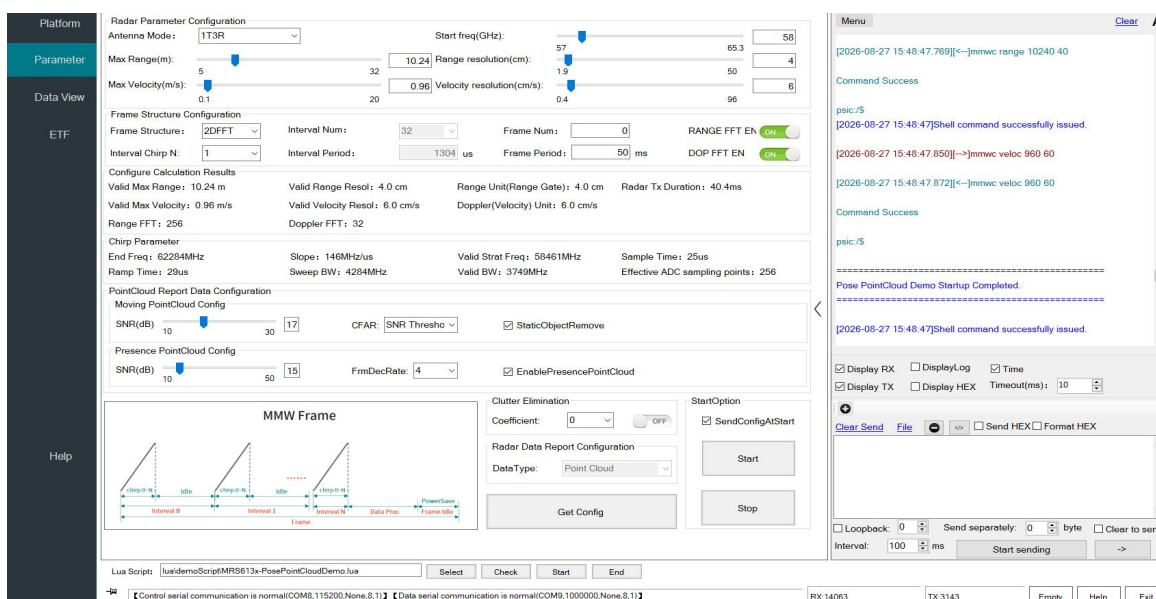


Figure 2-16 Presence Point Cloud Default Configuration Page

2.2.3 Motion Point Cloud/Tracking Display

After connecting to the Device as described in Section 2.1, perform the following steps:

1. Click Motion Point Cloud/Tracking Display to start the motion-point-cloud/tracking demo.
2. The software automatically switches to Data View, as shown in Figure 2-17. For details, see Data View Display (Point Cloud).
3. This display uses a fixed one-click radar configuration that cannot be modified. The configuration is shown in Figure 2-18.
4. Note: To perform another operation or change the configuration while the radar is running, first click Stop at the bottom of the Parameter Configuration or Data View page.

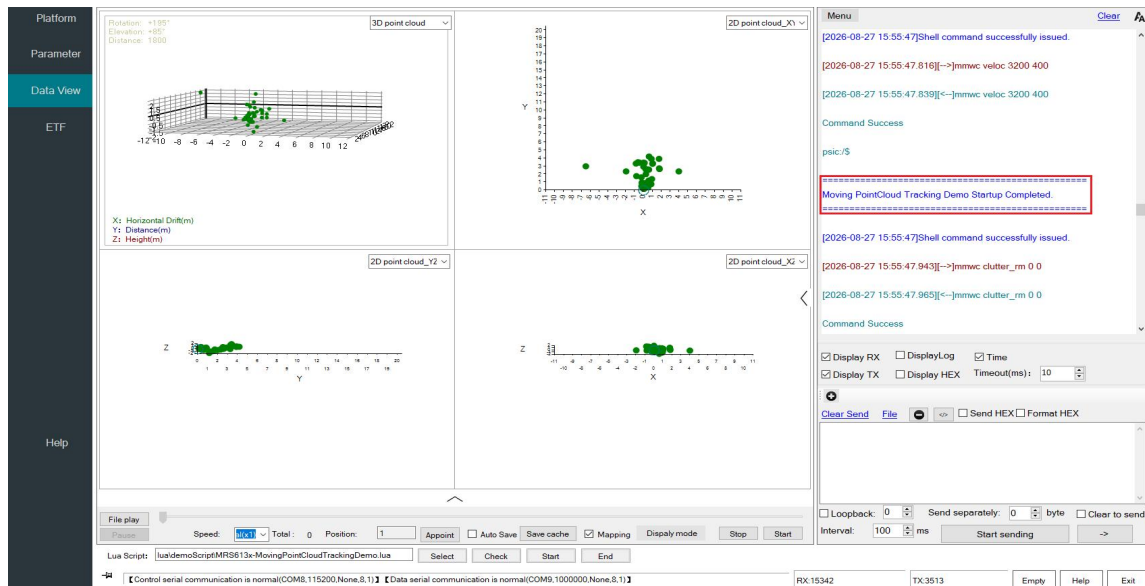


Figure 2-17 Motion Point Cloud/Tracking Data View

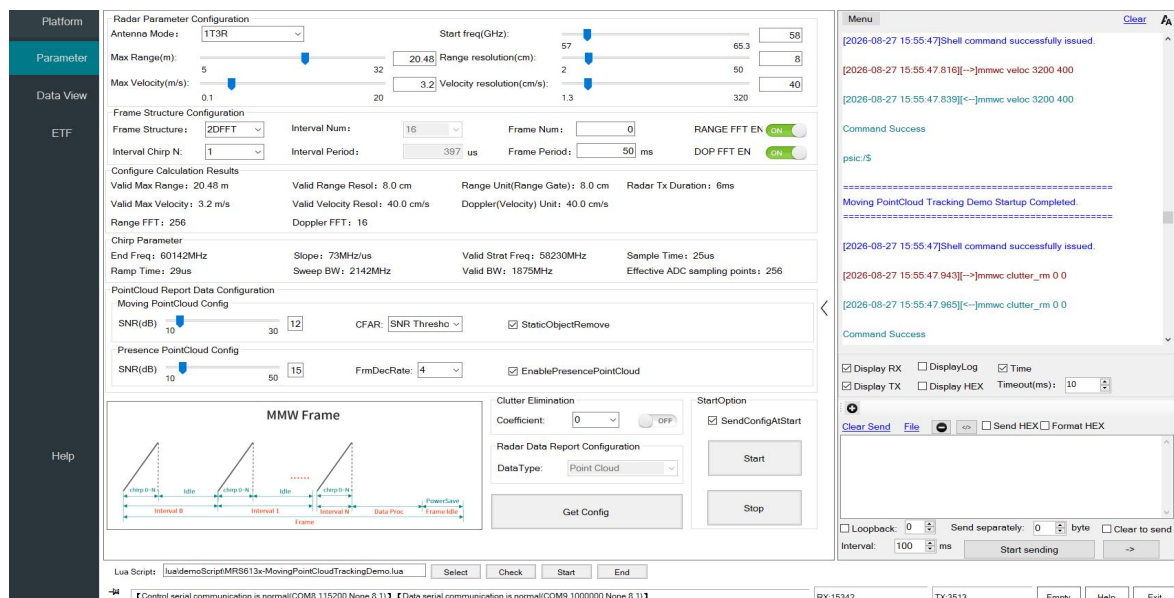


Figure 2-18 Motion Point Cloud/Tracking Default Configuration Page

2.3 Configuring Radar Operating Parameters

2.3.1 Configuration Flow

This section describes the radar-parameter configuration flow and explains the applicable parameters.

The Parameter Configuration page allows you to modify the radar operating parameters. Figure 2-19 shows the configuration flow. Configure the parameters as follows:

- 1、 Configure radar parameters by dragging the sliders or entering values directly:
 - (1) Multi-Antenna Mode: Selects the enabled antenna configuration. 1T1R indicates one transmit antenna and one receive antenna. 1T3R indicates one transmit antenna and three receive antennas.
 - (2) Start Frequency (GHz): Radar carrier frequency. The default is 58 GHz.
 - (3) Maximum Detection Range (m): Maximum unambiguous detection range of the radar.
 - (4) Range Resolution (cm): Minimum separation that the radar can resolve. Multiple targets separated by less than this value are resolved as one target.
 - (5) Maximum Detectable Velocity (m/s): Maximum unambiguous velocity detectable by the radar. The velocity measurement range is [-Maximum Detectable Velocity, Maximum Detectable Velocity]. Targets beyond this range are measured incorrectly.
 - (6) Velocity Resolution (cm/s): Minimum velocity difference that the radar can resolve. Multiple targets with a velocity difference below this value are resolved as one target.

Range and velocity FFT-size limits: The maximum Range FFT size is 1024. Range FFT size = Maximum Detection Range / Range Resolution, rounded up to the next power of two.

The maximum Doppler FFT size is 1024. Doppler FFT size = Maximum Detectable Velocity * 2 / Velocity Resolution, rounded up to the next power of two.

For best results, configure the parameters so that $\log_2(\text{Maximum Detection Range} / \text{Range Resolution})$ and $\log_2(2 * \text{Maximum Detectable Velocity} / \text{Velocity Resolution})$ are integers. Otherwise, spectral leakage can affect the results.
- 2、 Configure the frame structure according to the Frame Structure diagram. For details about the diagram, see Section 2.3.2.
 - (1) Frame Structure: In 1D FFT mode, a 1D FFT is performed in the range dimension and the radar reports target range information. In 2D FFT mode, a 1D FFT is performed in the range dimension followed by a 1D FFT in the Doppler dimension, and the radar reports target range and velocity information.
 - (2) 2D FFT Order (Interval N): Number of Intervals used for the Doppler FFT. The software calculates this value automatically from Maximum Detectable Velocity and Velocity Resolution.
 - (3) Number of Frames: Maximum number of frames captured continuously. Set this value to 0 for unlimited capture.
 - (4) Interval Chirp N: Number of chirps inserted in each Interval of the frame structure. An excessively large value can cause data errors and the abnormal spectrum shown in Figure 2-20. If this occurs, reduce the configured number of chirps.
 - (5) Interval Period: Time between two Intervals, including the up-ramp time of N chirps and the Idle time between Intervals. The software calculates this value automatically.
 - (6) Frame Period: Configurable time between two Frames, including N Intervals, DataProc, and Frame Idle. Set this value greater than the Frame period calculated automatically by the software; otherwise, the calculated period is used as the effective Frame period.
 - (7) RANGE FFT EN: Enables or disables the range-dimension FFT.
 - (8) DOP FFT EN: Enables or disables the Doppler-dimension FFT.

Note: If both Range FFT and Doppler FFT are disabled, ADC data is output.
- 3、 Radar Parameter Calculation Results and Chirp Parameters show the effective radar parameters derived from the preceding settings.
- 4、 Configure the reported point-cloud data by dragging the sliders, entering values directly, and selecting the applicable check boxes, including Static Target Removal and Enable Presence Point Cloud. These settings apply only when Reported Data Type is set to Point Cloud.
 - (1) Motion Point Cloud Configuration:
 - SNR (dB) (10 to 30): SNR threshold used to further filter motion points detected from Point Cloud data by the CFAR algorithm.
 - CFAR Algorithm: CA-CFAR uses an adaptive threshold for target detection; SNR Threshold uses a fixed threshold.

- Static Target Removal: Disabled by default. When enabled, removes the zero-Doppler bin and adjacent ± 1 low-frequency bins for targets stationary relative to the radar. When disabled, target points from all Doppler bins are reported.
- (2) Presence Point Cloud Configuration:
- SNR (dB) (10 to 50): SNR threshold for filtering presence-point-cloud information in Point Cloud data. The default is 15 dB.
 - Frame Extraction Rate: The default is 4. One frame is extracted from the DataCube every four frames to form a Micro DataCube. For more information, see Section 4.2.4.5, Presence Point Cloud, in the RS6x_7x mmWave Radar Sample Reference Manual.
 - Enable Presence Point Cloud: Enabled by default. Controls whether presence-point-cloud processing and Report are enabled. This setting is associated with the display described in Section 2.2.2.
- 5、 Select whether to enable Clutter Removal. When enabled, low-frequency energy in the acquired radar data is suppressed in real time, as shown in Figure 2-21 and Figure 2-22. The clutter-removal coefficient ranges from 0 to 15. A larger value produces slower convergence of static-clutter updates.
- 6、 Select the radar Reported Data Type.
- (1) DataCube: The radar reports range data or range-Doppler data according to the FFT dimensions selected by Frame Structure. The Data View is associated with the 2D DataCube Display described in Section 2.2.1.
- (2) Point Cloud: The radar reports target range, velocity, SNR, angle, and other data. The Data View is associated with the Presence Point Cloud Display in Section 2.2.2 and the Motion Point Cloud/Tracking Display in Section 2.2.3.
- 7、 Click Start.
- 8、 Switch to Data View to inspect the reported data.

The screenshot displays the POSSUMIC radar configuration software interface. On the left, a vertical sidebar contains buttons for 'Platform', 'Parameter', 'Data View', 'ETF', and 'Help'. The 'Data View' button is highlighted with a red box, and a red arrow points from it to the 'ETF' section. The main area is titled 'Radar Parameter Configuration 1. Config Radar param' and includes sliders and dropdowns for 'Antenna Mode', 'Max Range', 'Max Velocity', 'Start freq', 'Range resolution', and 'Velocity resolution'. Below this is 'Frame Structure Configuration 2. Config radar frame param' with settings for 'Frame Structure', 'Interval Num', 'Frame Num', 'Interval Chirp N', 'Interval Period', and 'Frame Period'. The 'Configure Calculation Results' section shows calculated values for range and velocity resolution. The 'Chirp Parameter 3. Actual effective configuration parameters for radar' section lists radar-specific parameters. The 'PointCloud Report Data Configuration 4. Config point cloud param' section includes 'Moving PointCloud Config' and 'Presence PointCloud Config' with SNR thresholds and CFAR settings. The 'Clutter Elimination 5. Config clutter remove' section features a coefficient slider and a 'StartOption' dropdown. The 'Radar Data Report Configuration' section shows 'Data Type' set to 'DataCube' and a 'Startup radar Start' button. At the bottom, there are 'Get Config' and 'Stop' buttons. A diagram of an 'MMW Frame' is shown, illustrating the sequence of chirps, idle time, and intervals.

Figure 2-19 Parameter Configuration Page

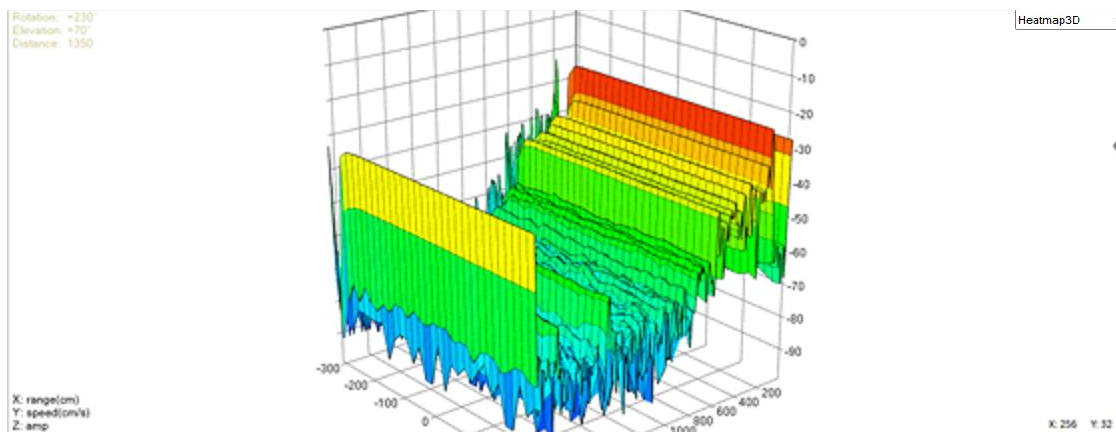


Figure 2-20 Abnormal Spectrum Caused by an Excessive Interval Chirp N

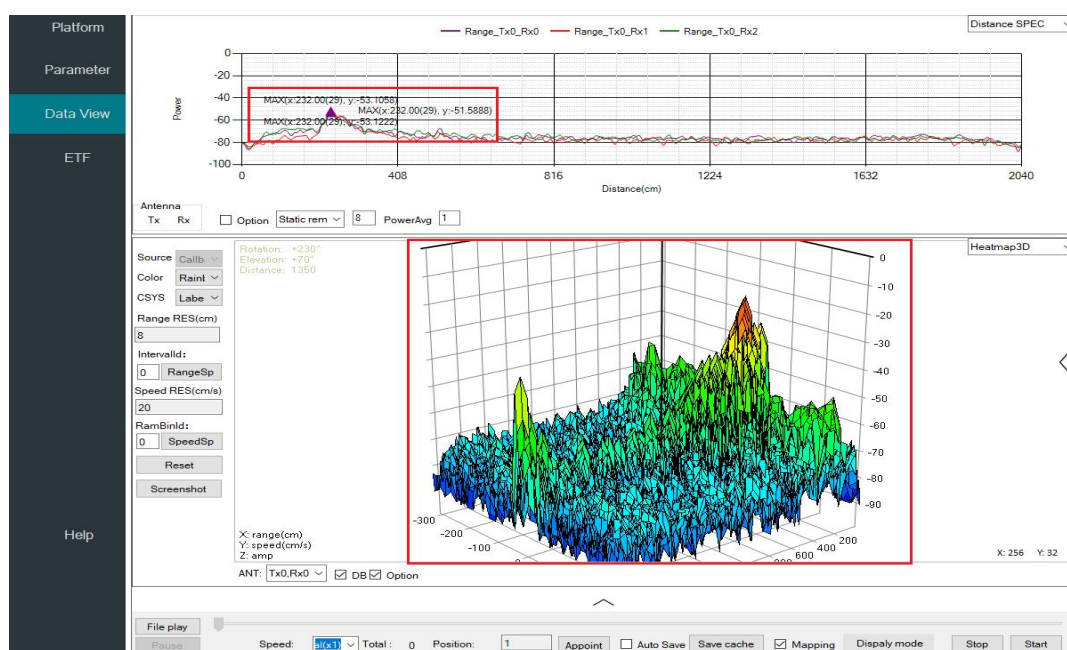


Figure 2-21 Before Static Removal Is Enabled

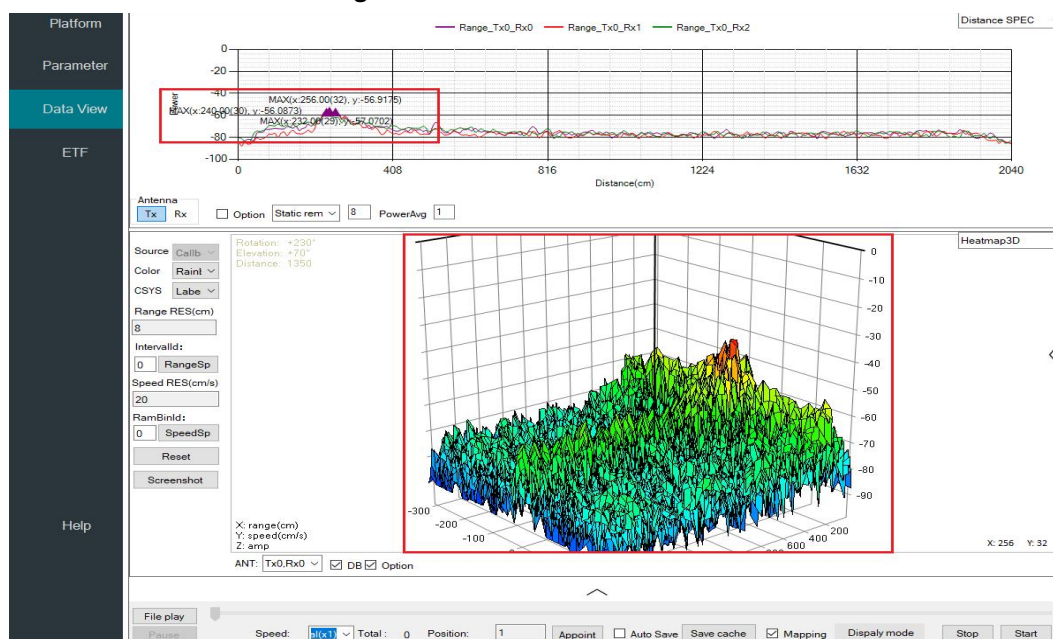


Figure 2-22 After Static Removal Is Enabled

2.3.2 FMCW Frame Structure

An FMCW frame consists primarily of a Chirp structure and a Frame structure. The Chirp structure is shown in Figure 2-23, and its parameters are described in Table 2-2.

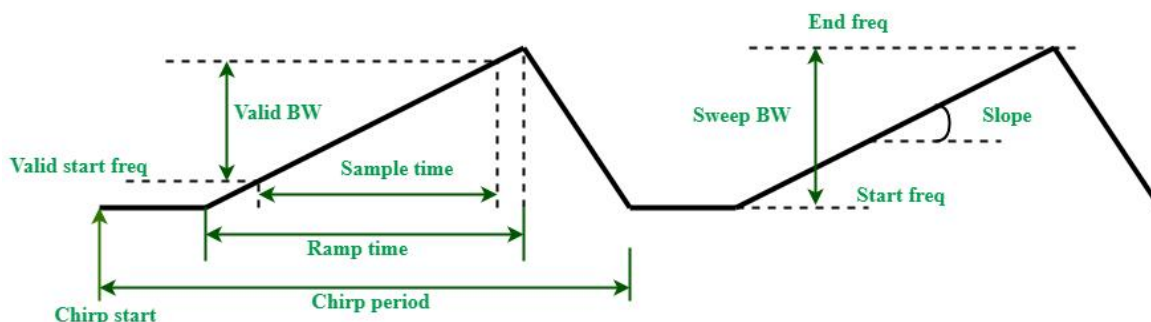


Figure 2-23 Chirp Structure

Table 2-2 Chirp Structure

Chirp Parameters	Description
Start Freq	Start Frequency F_s : Lowest frequency in the sweep. The user configures the start frequency from 57 GHz to 65 GHz.
End Freq	End Frequency F_e : Highest frequency in the sweep, up to 70 GHz. $F_e = F_s + BW$, where F_s is the start frequency and BW is the sweep bandwidth.
Slope	Sweep Slope S : Up-chirp sweep rate. $S = BWI * BWv / (d/r)$, where BWI is the fixed 10-MHz IF sampling rate, BWv is the effective sweep bandwidth, d is the maximum detection range, and r is the range resolution.
Sweep BW	Sweep Bandwidth BW : $BW = BWv + S * 3.65$, where BWv is the effective sweep bandwidth and S is the sweep slope.
Valid Start Freq	Effective Start Frequency F_{sv} : Start frequency of ADC sampling. $F_{sv} = F_s + 3.15 * S$, where F_s is the start frequency and S is the sweep slope.
Valid BW	Effective Sweep Bandwidth BWv : Bandwidth from the start to the end of ADC sampling. $BWv = c/(2r)$, where c is the speed of light and r is the range resolution.
Sample Time	Effective Sampling Time T_s : Valid ADC sampling time during the up-ramp. $T_s = BWv/S$, where BWv is the effective sweep bandwidth and S is the sweep slope.
Sample Number	Number of Valid ADC Samples N : This value is not necessarily equal to the FFT size. If the number of ADC output samples is smaller than the FFT size, a zero-padded FFT is performed. $N = d/r$, where d is the maximum range and r is the range resolution.
Ramp Time	Up-Ramp Time T_r : Time required to ramp from Start Frequency to End Frequency. $T_r = BW/S$, where BW is the sweep bandwidth and S is the sweep slope.
Chirp period	Chirp Period T_c : Time from the start of one Chirp to the start of the next Chirp in the same Interval. $T_c = \text{hardware power-up/power-down time} + T_r$, where T_r is the up-ramp time.

Using the RS6130 as an example, Figure 2-24 shows the FMCW Frame structure and Table 2-3 describes its parameters.

Frame Structure

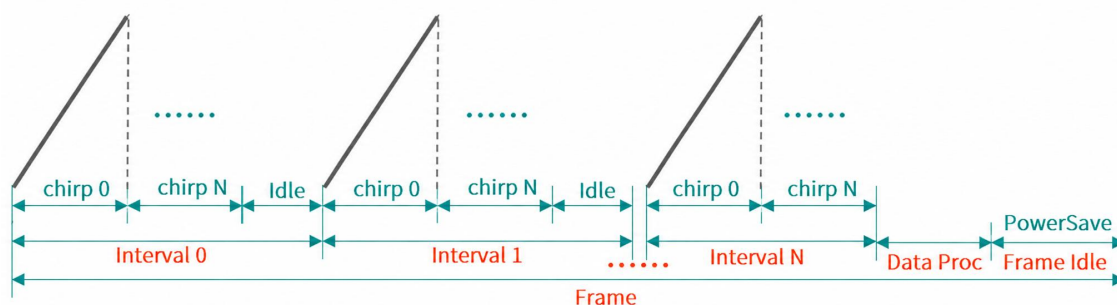


Figure 2-24 Frame Structure

Table 2-3 Frame Structure Parameters

Frame Parameter	Description
Frame	A complete radar detection cycle is called a Frame. Frame Period is the radar detection period, and Number of Frames is the number of detection cycles to perform.
Interval	A Frame consists of one or more measurement cycles called measurement Intervals. Chirps are transmitted and Range FFT processing is completed within a measurement Interval, which is the basic ranging unit. In 2D FFT mode, Doppler FFT processing is performed across multiple measurement Intervals. Range FFT and Doppler FFT are performed independently for each TX/RX channel, such as RX0, RX1, and RX2. A Frame can end with one or more idle Intervals. No Chirp is transmitted during an Idle Interval, which can be used for frame data processing, waiting, or sleep.
Chirp N	One or more Chirps can be transmitted continuously on a TX channel within an Interval. Multiple Chirps are contiguous and are integrated before Range FFT processing. The default number of Chirps is currently 1.
Data Process	After the Range FFT for the last measurement Interval in a Frame is complete, the system enters the data-processing stage. This stage includes Doppler FFT processing, CFAR detection, AoA processing, and triggering upper-layer data processing.
Frame Idle	After data processing is complete, the radar module enters the idle state. If power-saving mode is enabled, the radar system enters the power-saving state. <i>Downstream application-layer processing must be completed before Frame Idle ends; otherwise, a data-access conflict or error can occur.</i>

2.4 Data View

This section describes how Data View visualizes radar Report data. For DataCube, the available views are Range Spectrum, Doppler Spectrum, 2D and 3D Range-Doppler Heatmaps, and IQ Plot. For Point Cloud, the available views are 2D and 3D point-cloud plots.

2.4.1 Data View Display (DataCube)

The Data View page is shown in Figure 2-25.

After connecting to the Device as described in Section 2.1, switch to Data View and use the following controls:

- 1、 Under Display Layout, select a single-view, two-view, or four-view layout.
- 2、 As shown in Figure 2-26, Data View can display data from a specified antenna to facilitate debugging.

3、 As shown in Figure 2-27, select Configuration Options to open the configuration pane. This pane provides curve configuration, resolution settings, and signal analysis.

4、 As shown in Figure 2-28, the Range Spectrum view provides Static Removal, Single Frame, and Smoothing processing options.

Assume that the Range FFT size is N_{RFFT} and the Doppler FFT size is N_{DFFT} .

- (1) Static Removal: Effective only when Frame Structure is configured for 2D FFT. The power of each range-Doppler bin in the DataCube is calculated, and Doppler bins $[N_{\text{DFFT}}/2-k, N_{\text{DFFT}}/2+k]$ are removed, where k is configurable. Press Enter to apply the setting. The result is then averaged along the Doppler dimension to reduce interference from stationary or low-velocity targets and emphasize moving targets. This calculation is performed for every channel, as shown in Figure 2-30.
- (2) Single Frame: Calculates the power of all range bins at zero velocity and plots the result. The calculation flow is shown in Figure 2-31.
- (3) Smoothing: Based on the Single Frame result, calculates the mean of the most recent k frames, where k is configurable, to reduce random-noise fluctuations and improve target-detection stability. The calculation flow is shown in Figure 2-32.
- (4) Interframe Subtraction: Effective for both 1D FFT and 2D FFT Frame structures. In 1D FFT mode, only one Interval of data is reported per frame. The Interval data from adjacent frames is subtracted before power is calculated to emphasize moving targets, as shown in Figure 2-33. In 2D FFT mode, the zero-velocity Doppler data is extracted from each frame. The zero-velocity data from adjacent frames is subtracted before power is calculated to detect moving targets, as shown in Figure 2-34.
- (5) Intraframe Subtraction: Effective only when Frame Structure is configured for 2D FFT. The first two Intervals in each frame of the DataCube are subtracted before power is calculated, as shown in Figure 2-35.

5、 Doppler Spectrum: Calculates the power of each range-Doppler bin in the DataCube and averages the power of each Doppler bin along the range dimension, as shown in Figure 2-37.

6、 As shown in Figure 2-38, IQ Plot visualizes the zero-velocity component of a specified range bin. Three IQ data types are available: IQ Value Plot, IQ Phase Plot, and IQ Scatter Plot. The Range Bin text box selects a bin within the specified range. The NumericUpDown control to the right expands the displayed interval on both sides of the selected bin. Up to three range bins can be displayed simultaneously; therefore, NumericUpDown ranges from 0 to 1.

Assume that the IQ data is $a + bj$. The IQ Plot functions are defined as follows:

- (1) IQ Phase Plot: Displays $\text{atan}(b/a)$ in degrees.
- (2) IQ Value Plot: Displays the normalized IQ values $a/32768$ and $b/32768$.
- (3) IQ Scatter Plot: Uses the real component as the x-coordinate and the imaginary component as the y-coordinate to plot $(a+bj)/32768$ on the complex plane.

Note: Use the following controls to change the viewpoint in the 3D Range-Doppler Heatmap:

- (1) Elevation View: Hold down the left mouse button and move the mouse forward or backward.
- (2) Azimuth View: Hold down the right mouse button and move the mouse left or right.
- (3) Zoom In: Scroll the mouse wheel backward.
- (4) Zoom Out: Scroll the mouse wheel forward.

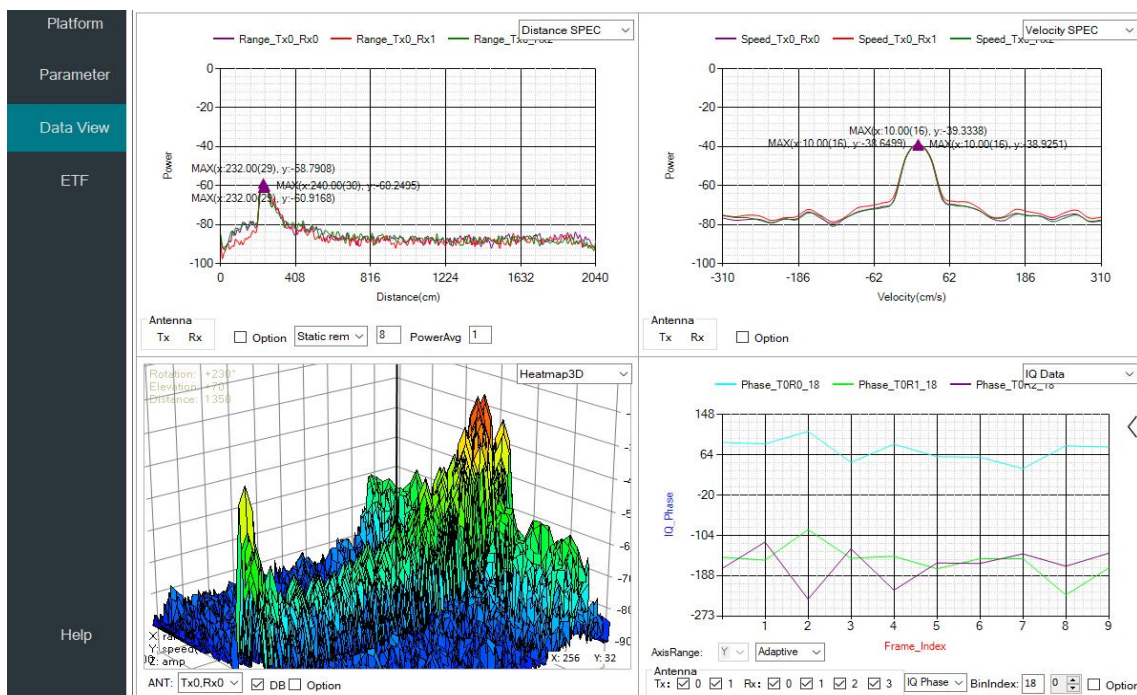


Figure 2-25 Data View Page

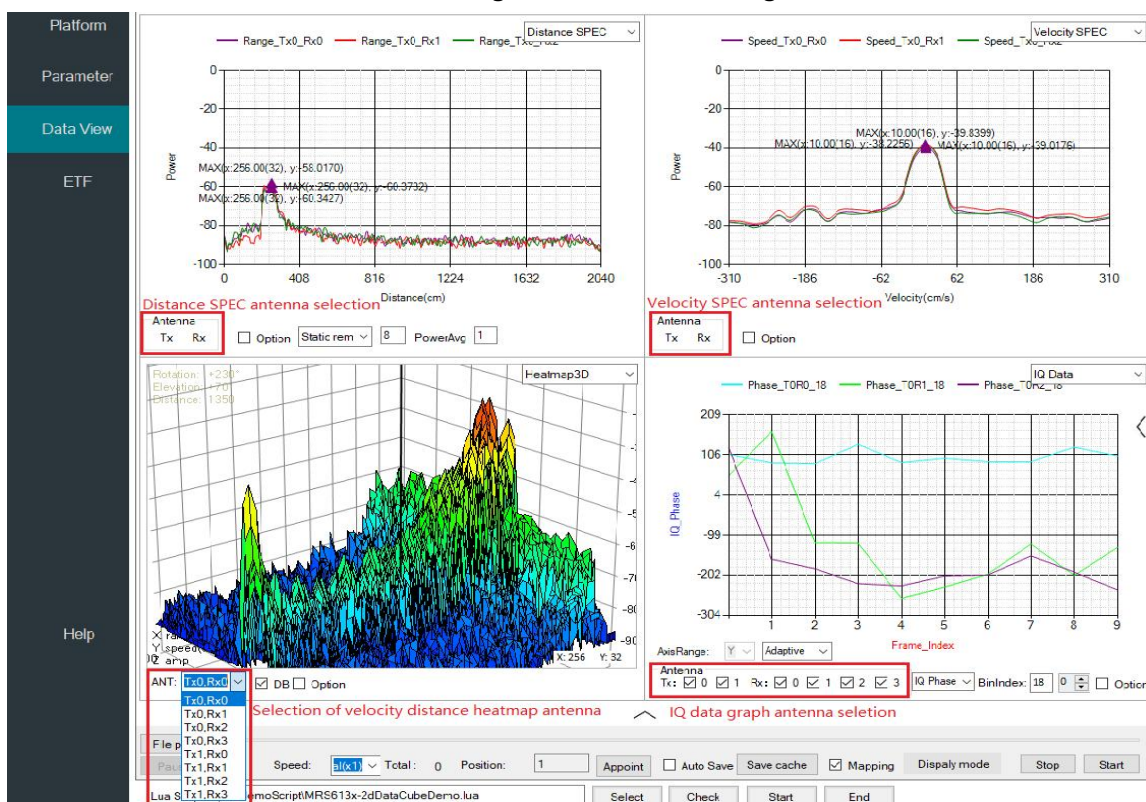


Figure 2-26 Antenna Selection

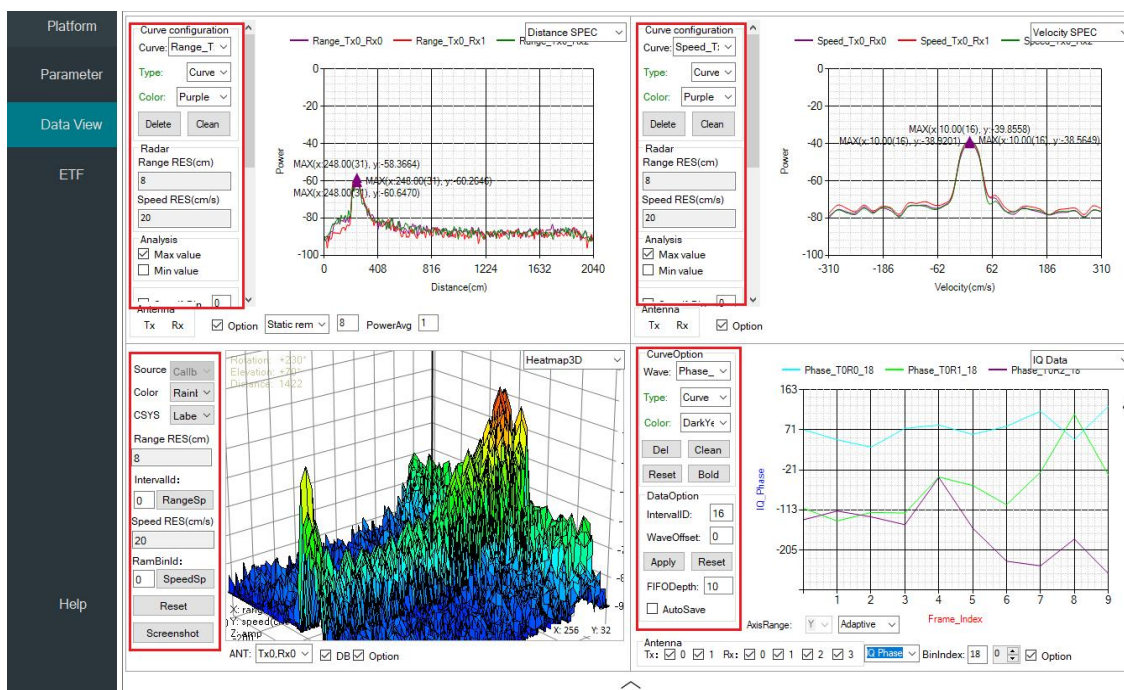


Figure 2-27 Configuration Options (DataCube)

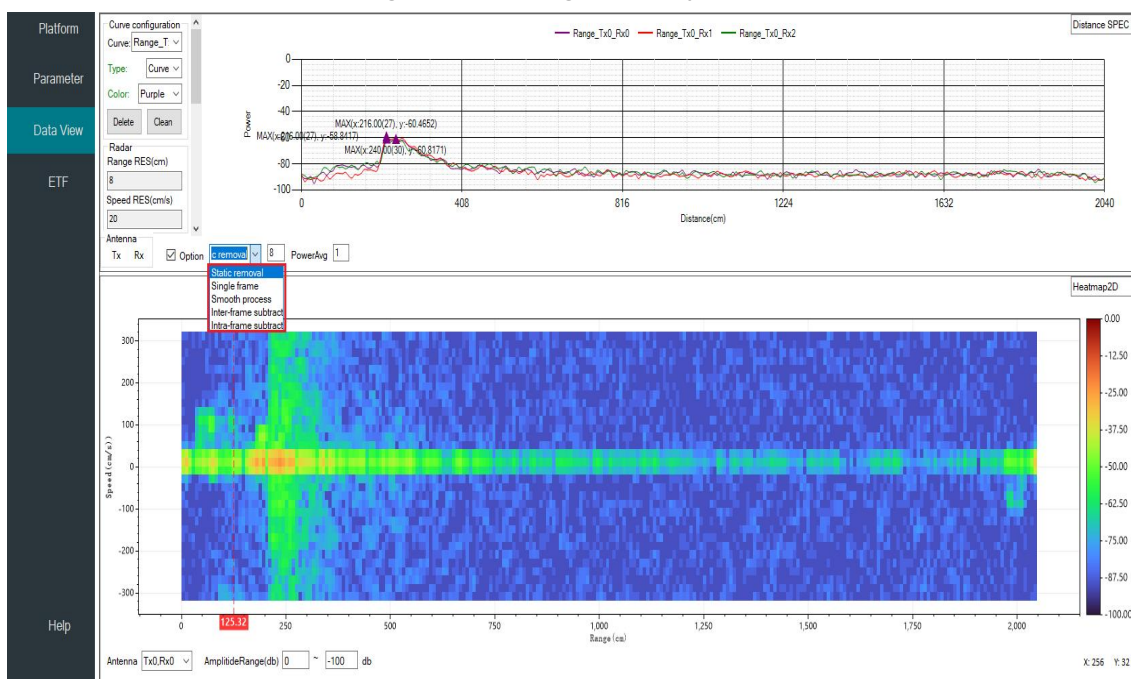
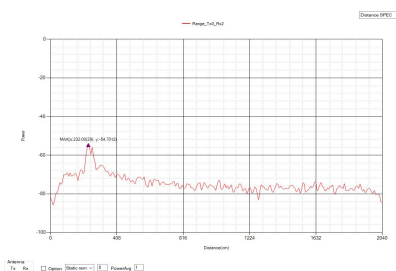
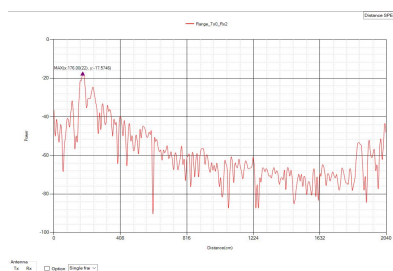


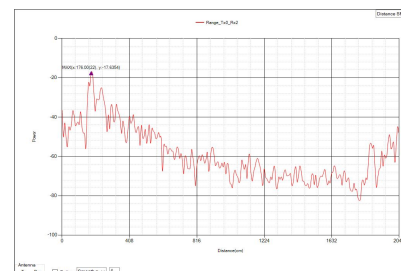
Figure 2-28 Range-Spectrum Function Selection



(a) Static Removal



(b) Single Frame



(c) Smoothing

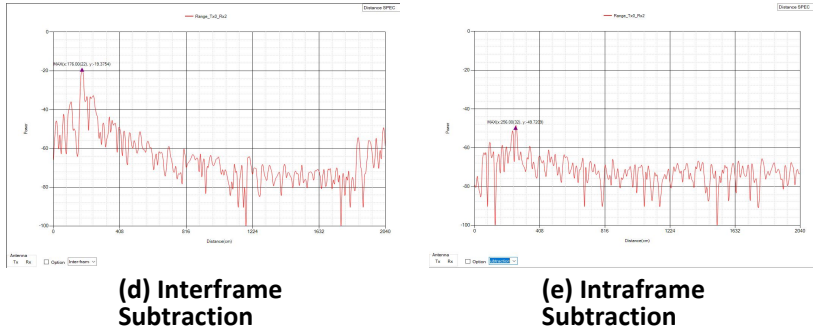


Figure 2-29 Range-Spectrum Data Processing

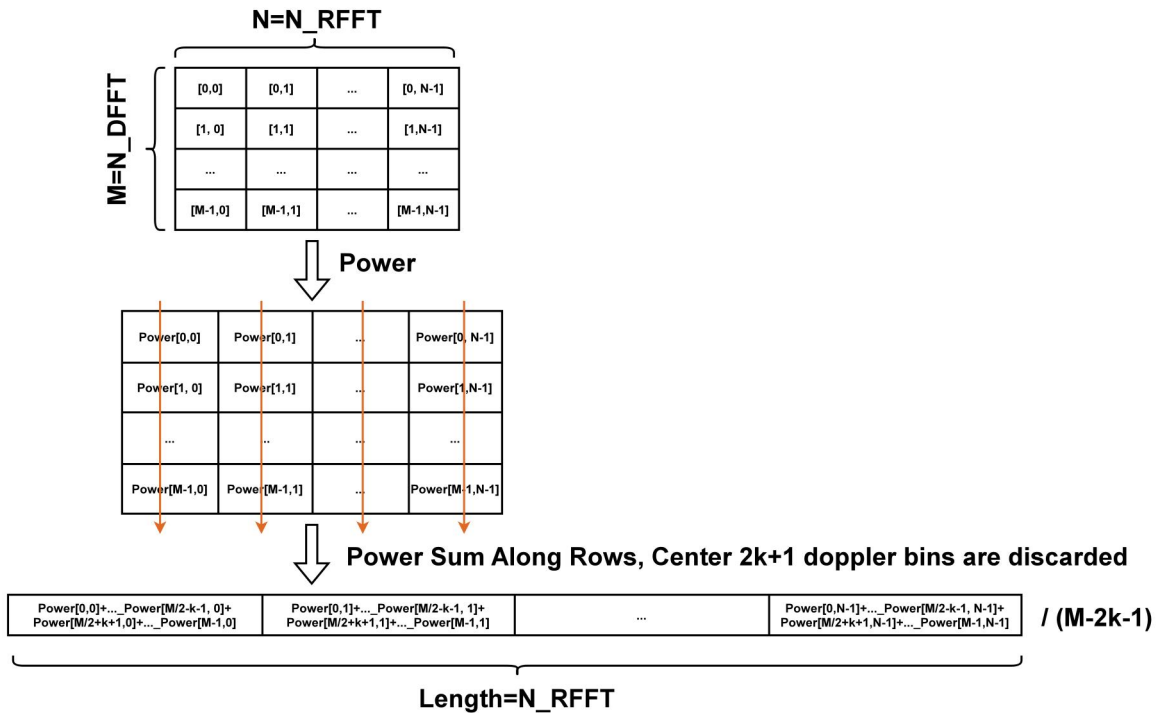


Figure 2-30 Static-Removal Calculation

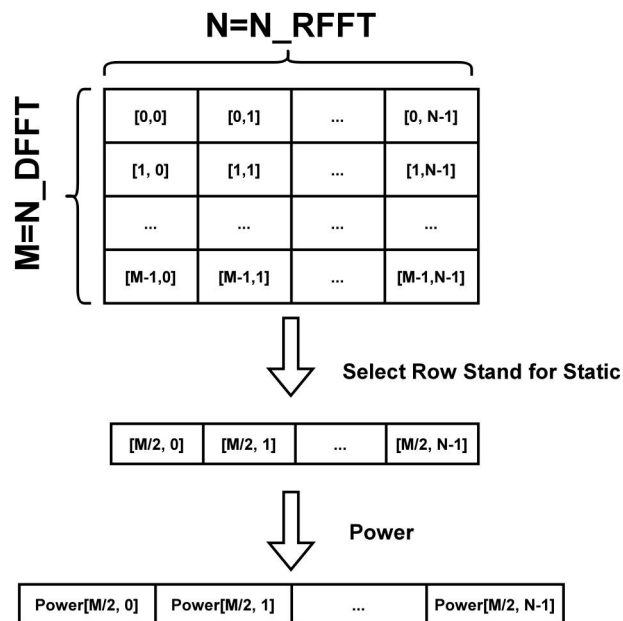


Figure 2-31 Single-Frame Calculation

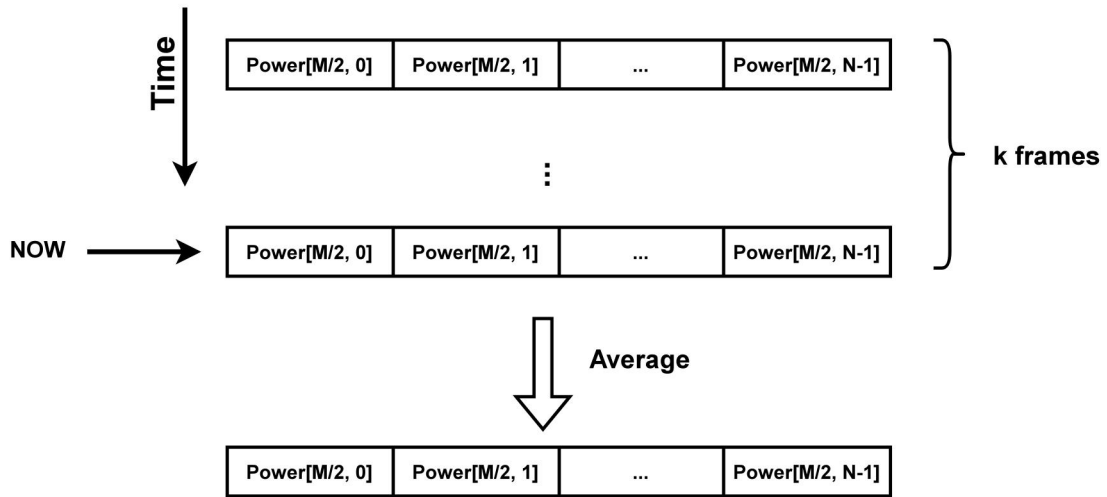


Figure 2-32 Smoothing Calculation

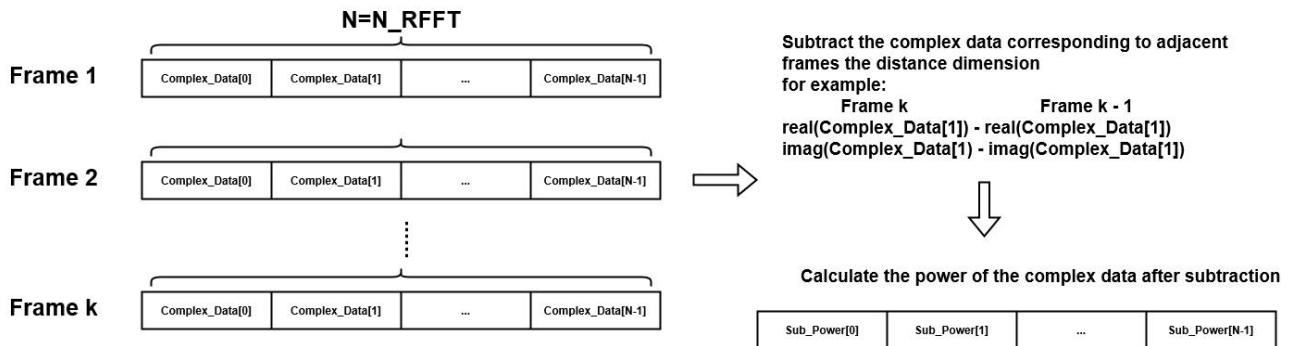


Figure 2-33 Interframe-Subtraction Calculation (1D FFT Frame Structure)

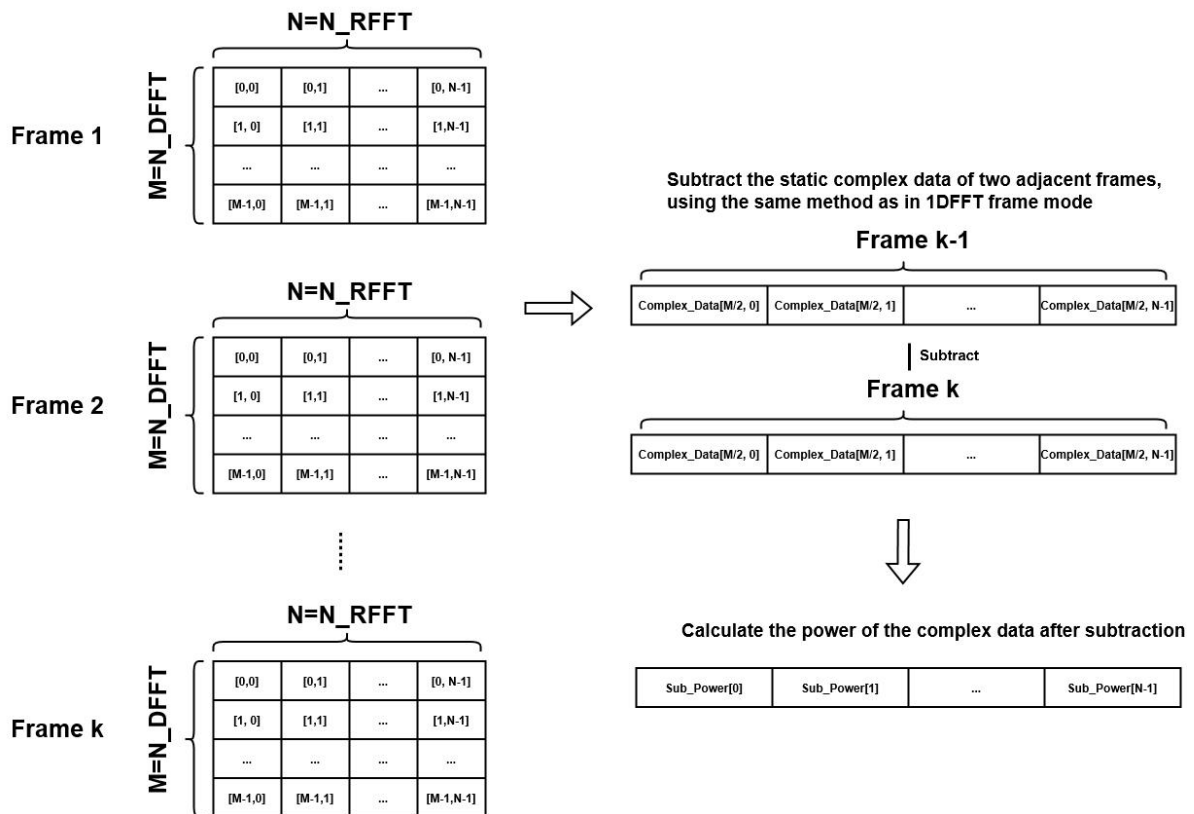


Figure 2-34 Interframe-Subtraction Calculation (2D FFT Frame Structure)

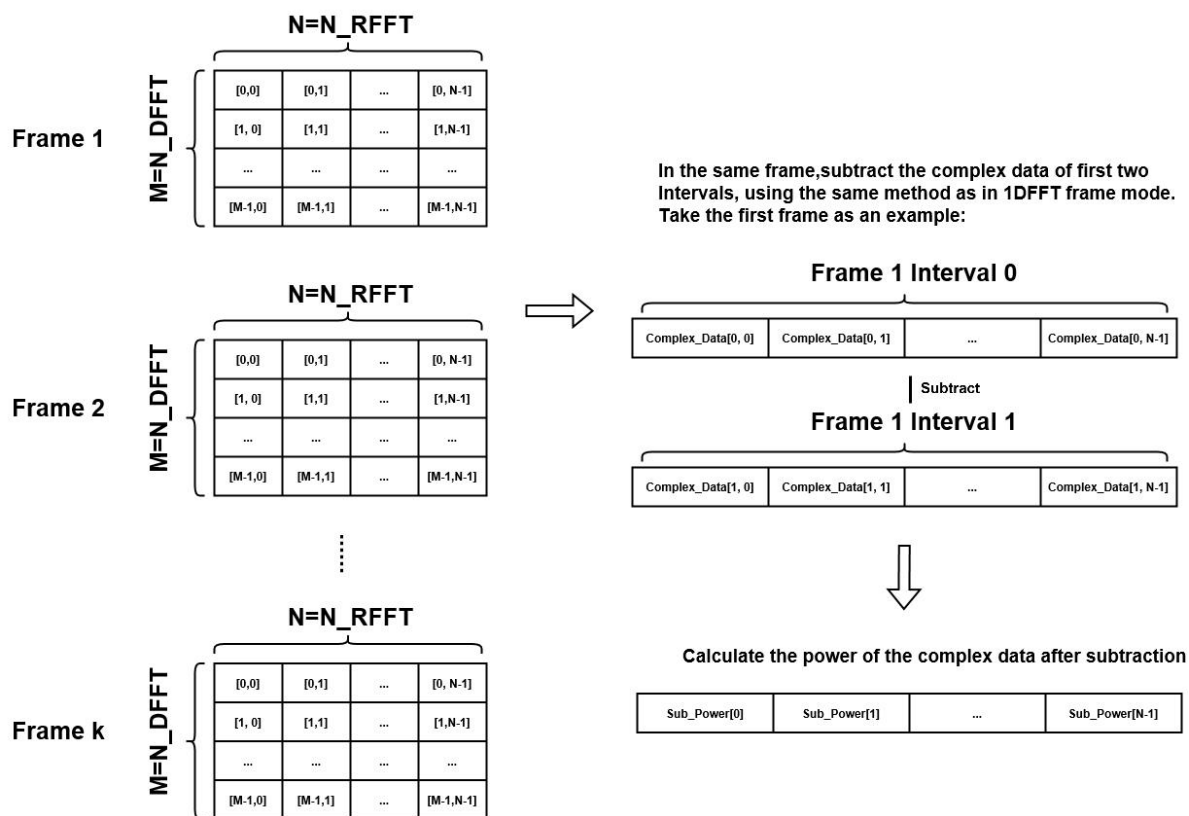


Figure 2-35 Intraframe-Subtraction Calculation

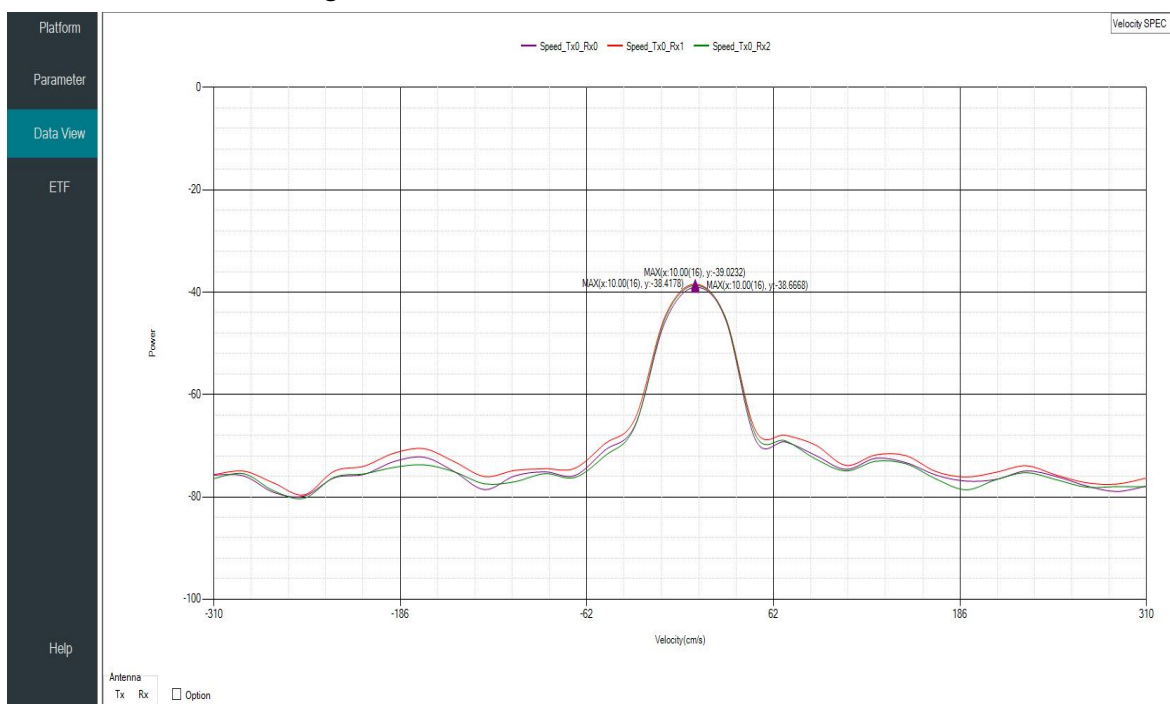


Figure 2-36 Doppler Spectrum

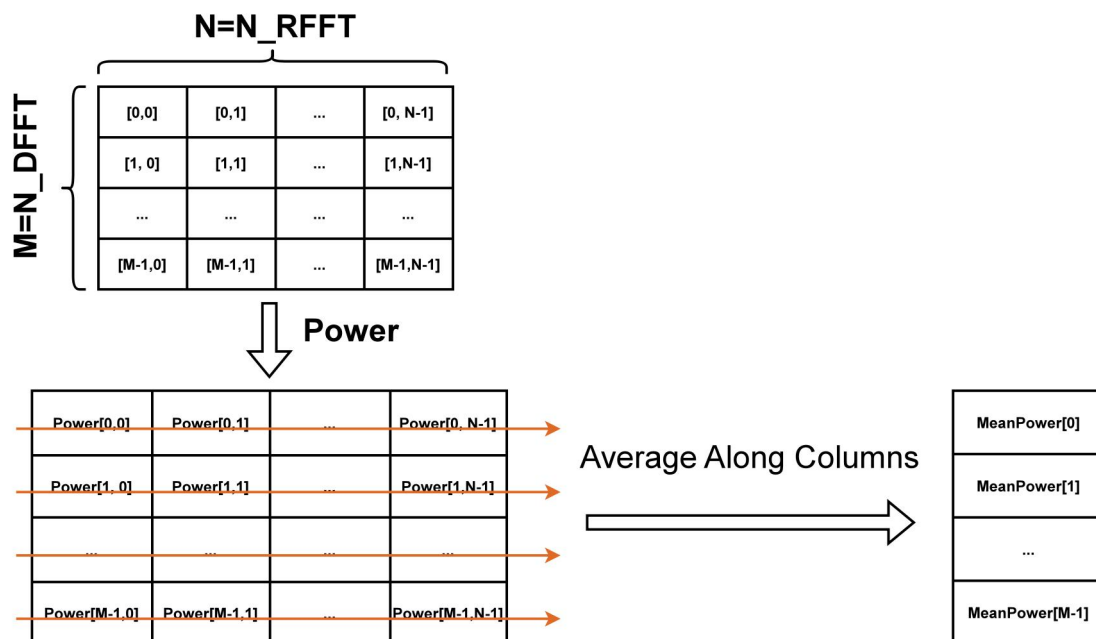


Figure 2-37 Doppler-Spectrum Calculation

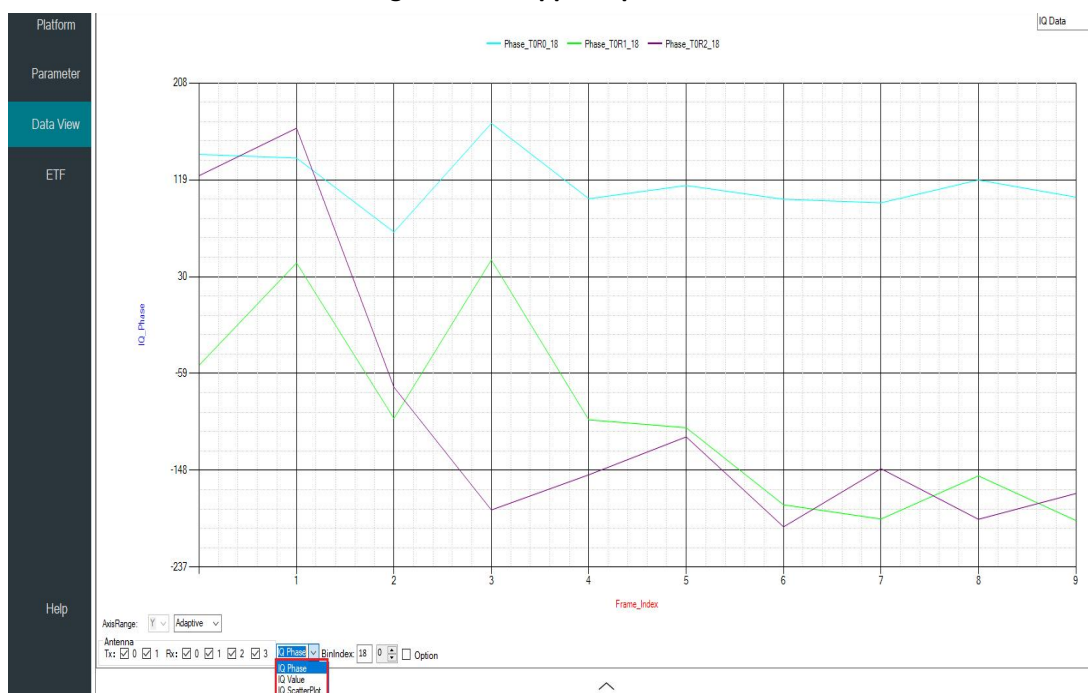


Figure 2-38 IQ Plot Data-Type Selection

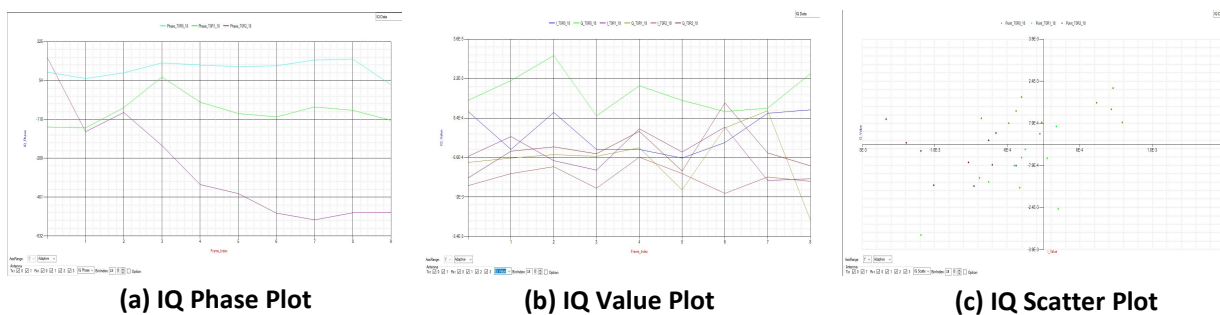


Figure 2-39 IQ Plot

2.4.2 Data View Display (Point Cloud)

After connecting to the Device as described in Section 2.1, switch to Data View and use the following controls:

- 1、 Under Display Layout, select the Data View layout: single-view, two-view, or four-view mode.
- 2、 As shown in Figure 2-40, right-click in each view to enable or disable Display Motion Point Cloud (blue, enabled by default), Display Presence Point Cloud (orange, enabled by default), and Other Configuration Options. In the example, presence point clouds are disabled in the 3D Point Cloud and 2D Point Cloud_XY views and enabled in the 2D Point Cloud_YZ and 2D Point Cloud_XZ views.
- 3、 As shown in Figure 2-41 and Figure 2-42, select Other Configuration Options to open the configuration panel. The panel provides coordinate-range adjustment (including axis inversion), display-range limits, accumulated-frame settings, point-cloud color selection, and other options.
- 4、 As shown in Figure 2-43, if continuous plotting causes the "Plotting is too slow. Plotting stopped." message to appear and plotting stops, reselect Enable Plotting at the bottom of the view.

Note: Use the following controls to adjust the 3D point-cloud view:

- (1) Elevation View: Hold down the left mouse button and move the mouse forward or backward.
- (2) Azimuth View: Hold down the right mouse button and move the mouse left or right.
- (3) Zoom In: Scroll the mouse wheel backward.
- (4) Zoom Out: Scroll the mouse wheel forward.

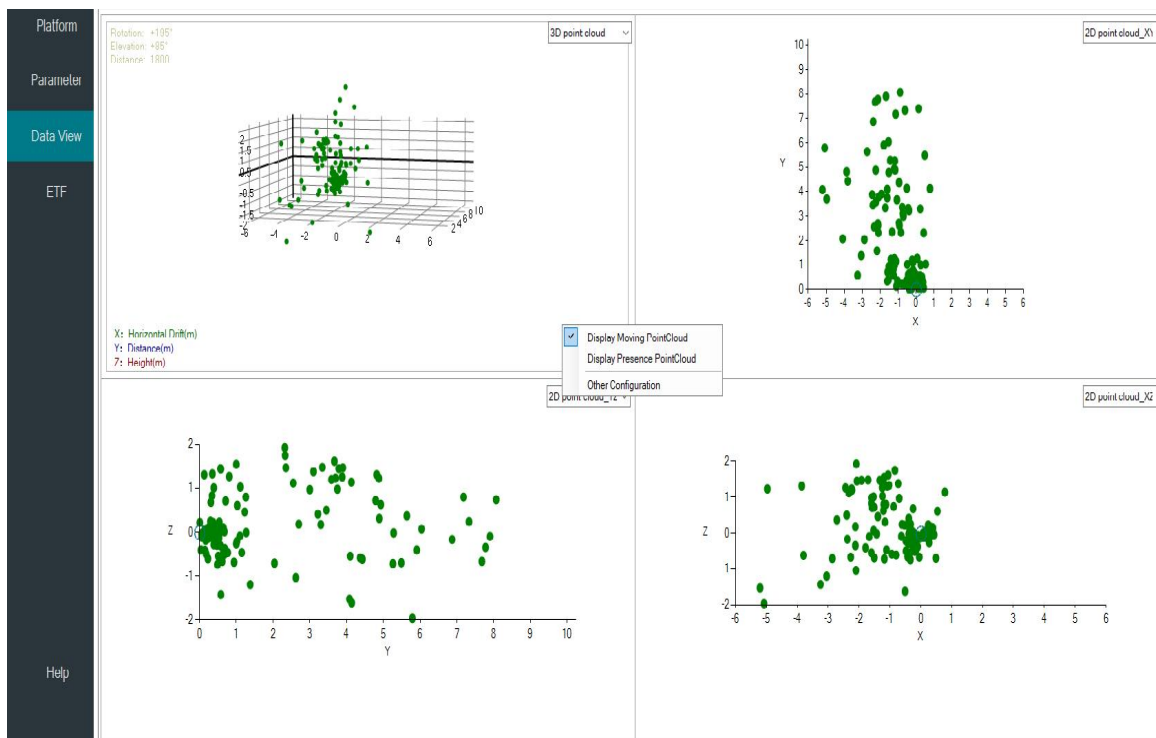


Figure 2-40 Point-Cloud View Selection Example

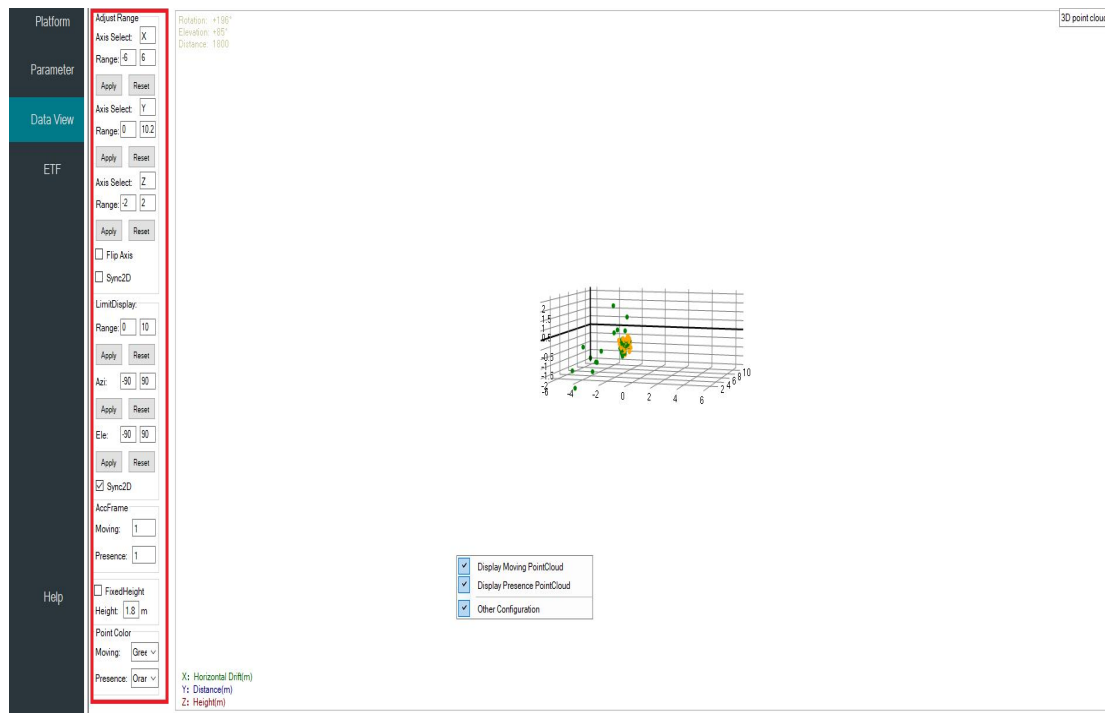


Figure 2-41 Configuration Page (3D Point Cloud)

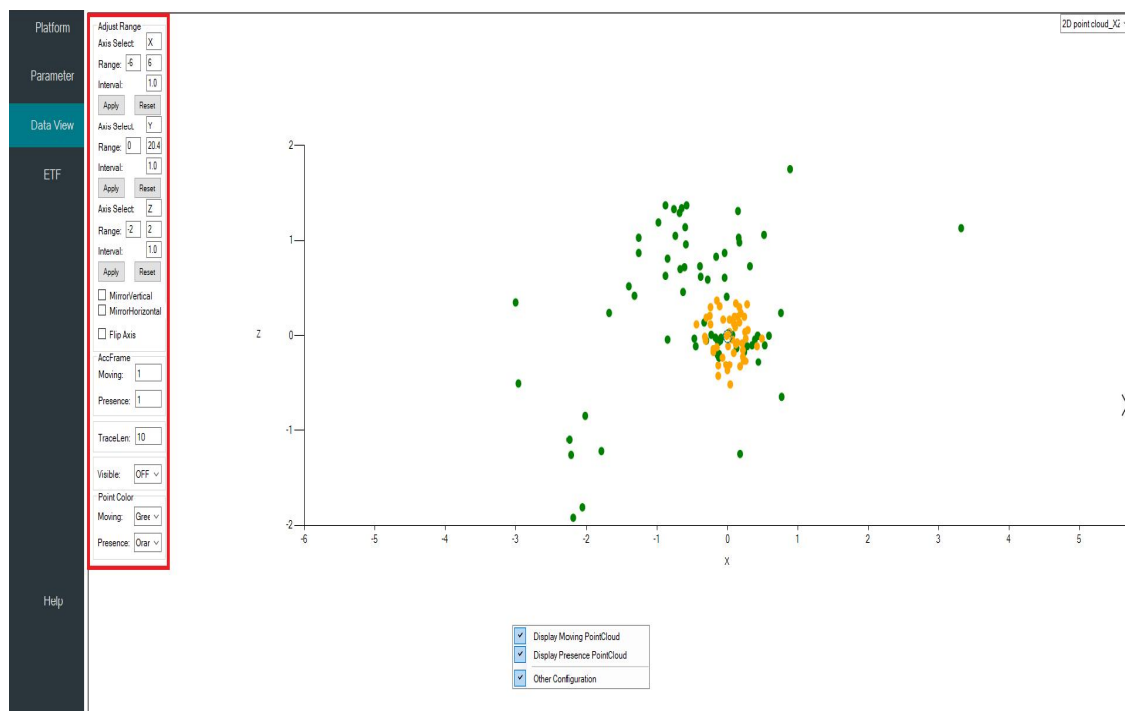


Figure 2-42 Configuration Page (2D Point Cloud)

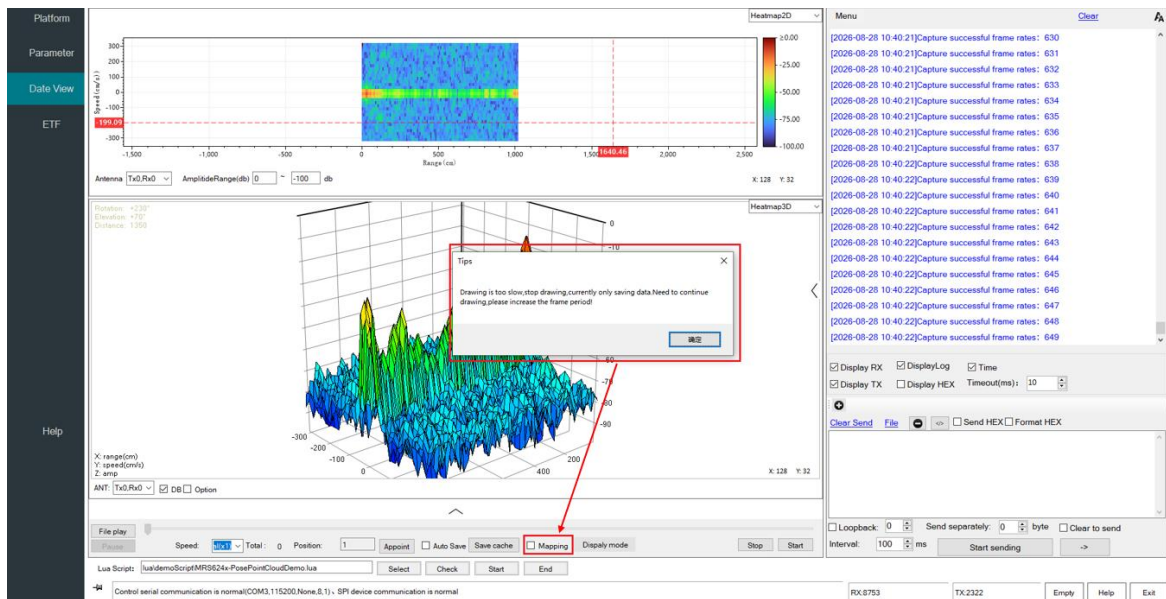


Figure 2-43 Restarting Plotting

2.5 Saving and Playing Files

This section describes the File Saving and File Playback functions for radar report data.

2.5.1 Saving a File

After connecting to the Device as described in Section 2.1, perform the following steps:

- 1、On the Data View page, select the Auto Save check box, select the save path, and specify a file name.
- 2、To save a fixed number of frames, modify Number of Frames under Parameter Configuration.
- 3、Start data capture. Data is saved automatically to the selected file.
- 4、If Stop is clicked during operation, the captured data is saved.

Note: Saved data files use the .dat extension. Two data types are available: DataCube and Point Cloud. The selected type is associated with Select the Radar Report Data Type in Section 2.3. To save files in additional formats, refer to Auto Save under Menu Functions in Section 2.8.1.

Note: A .dat file is used only for data playback. It stores serialized data and cannot be parsed directly. To parse a DataCube or point-cloud file, use Data Conversion in Section 2.8.2 to convert the .dat file to a .bin or .csv file.

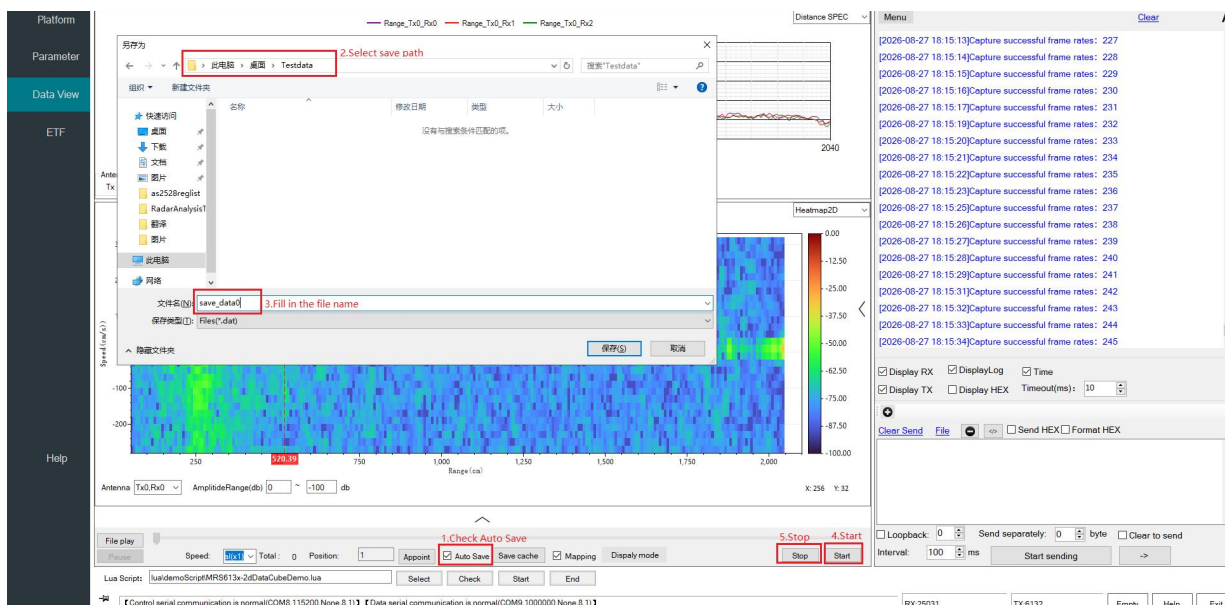


Figure 2-44 File Saving



Figure 2-45 File Saved Successfully

2.5.2 Playing a File

- 1、Click File Playback and select the .dat file to play back.
- 2、Click Open. The file data is automatically parsed and plotted.
- 3、During playback, click Pause Playback, enter the target frame in the Current Frame Position text box, and click Specified Frame or press Enter to jump directly to the specified frame.
- 4、Drag the progress bar with the left mouse button. Release the mouse button to jump directly to the specified frame.

Note: The data type selected for File Playback must match Select the Radar Report Data Type in Section 2.3; otherwise, playback fails.

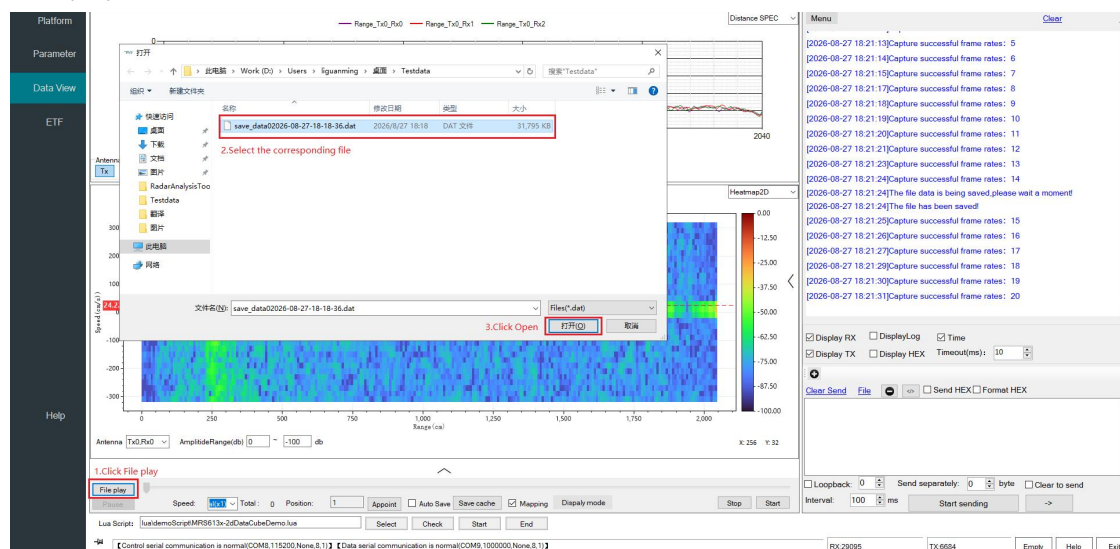


Figure 2-46 File Playback

2.6 ETF Debugging

This section describes radar debugging functions, including register access, static configuration, TX-channel debugging, and calibration.

2.6.1 Register Access

The Register Read/Write page is shown in Figure 2-47.

After connecting to the Device as described in Section 2.1, perform the following steps:

- 1、 Switch to the ETF page.
- 2、 Select Register Read/Write.
- 3、 Enter the register address and the value to write. For a read operation, no value is required.
- 4、 Click Read or Write.

Note: This function is generally used for temporary SDK and development debugging.

The screenshot shows the POSUMIC software interface with the 'ETF' tab selected in the left sidebar. The main window displays the 'Register operation' tab, which includes a 'Register operation' section. This section has three input fields: 'Address' (containing '0x40019040'), 'Value' (containing '0'), and 'Decimal Value' (empty). Below these fields is a 32-bit register value display, organized into two rows of 16 bits each. The top row shows bits 31 down to 16, and the bottom row shows bits 15 down to 0. All bits are currently set to '0'. To the right of the bit display are two buttons: 'Read' and 'Write'.

Figure 2-47 Register Read/Write Page

2.6.2 TX-Channel Debugging

The TX Channel Debugging page is shown in Figure 2-48.

After connecting to the Device as described in Section 2.1, perform the following steps:

- 1、 Switch to the ETF page.
- 2、 Select TX Channel Debugging.
- 3、 Enable TX-channel debugging.
- 4、 Configure the applicable parameters according to project requirements.
- 5、 After configuration, click Transmit Tone.

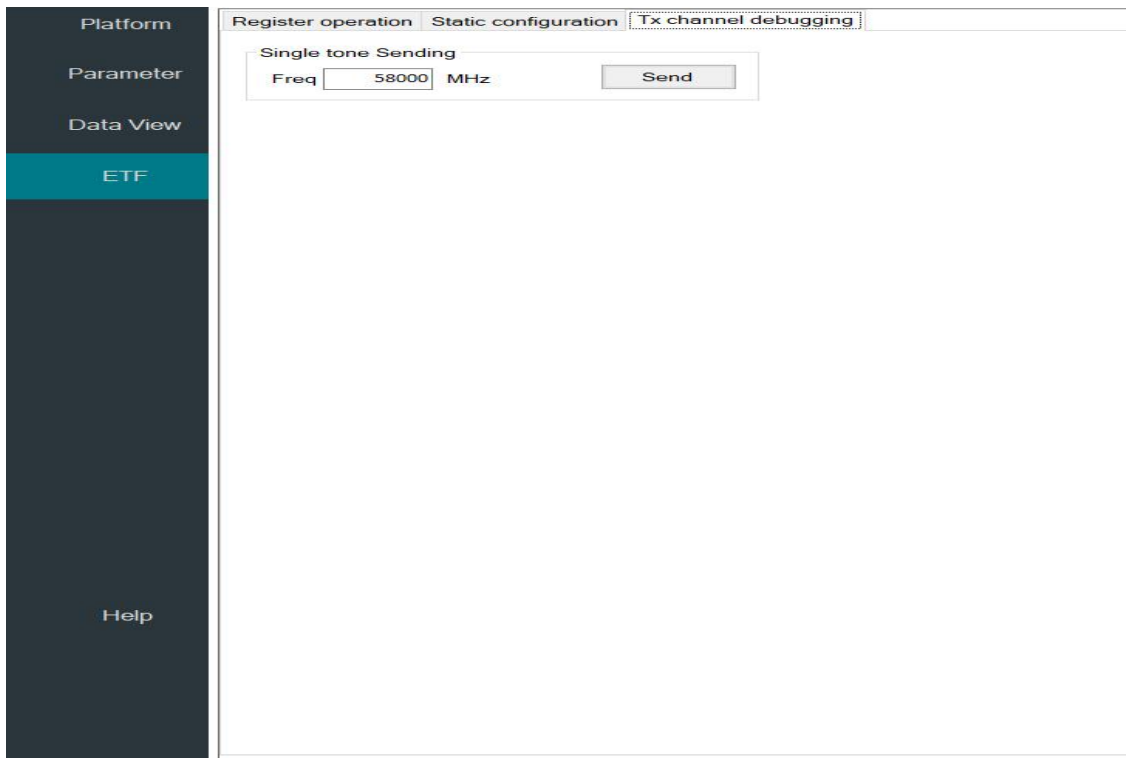


Figure 2-48 TX Channel Debugging Page

2.6.3 Static Configuration

The Static Configuration page is shown in Figure 2-49. This page provides global static settings.

After connecting to the Device as described in Section 2.1, perform the following steps:

- 1、 Switch to the ETF page.
- 2、 Select Static Configuration.
- 3、 Configure the applicable parameters according to project requirements. Recommended settings for typical scenarios are as follows:
 - (1) To detect weak targets at long range, use high RF gain and moderate BB gain, disable RF attenuation, and use a low HPF bandwidth.
 - (2) If a strong reflecting interferer is present at short range, reduce the RF and BB gains, enable RF attenuation, and use a high HPF bandwidth.

Note: The parameters are described below. Using an RF gain of 45 dB and an HPF bandwidth of 2 MHz as an example, Figure 2-50 shows the average range-spectrum signal power with the default settings (51 dB/0.5 MHz), and Figure 2-51 shows the result after the settings are changed to 45 dB/2 MHz.

- (1) RF Gain: Front-end amplifier gain, in dB. Increasing or decreasing this setting changes the overall signal power accordingly.
- (2) BB Gain: Baseband amplifier gain, in dB. Increasing or decreasing this setting changes the overall signal power accordingly.
- (3) RF Attenuation (dB): Adjustable attenuation in the transmit path. Increasing or decreasing this setting changes the overall signal power accordingly.
- (4) HPF Bandwidth: IF high-pass filter cutoff frequency, in MHz. This filter suppresses low-frequency signal components.

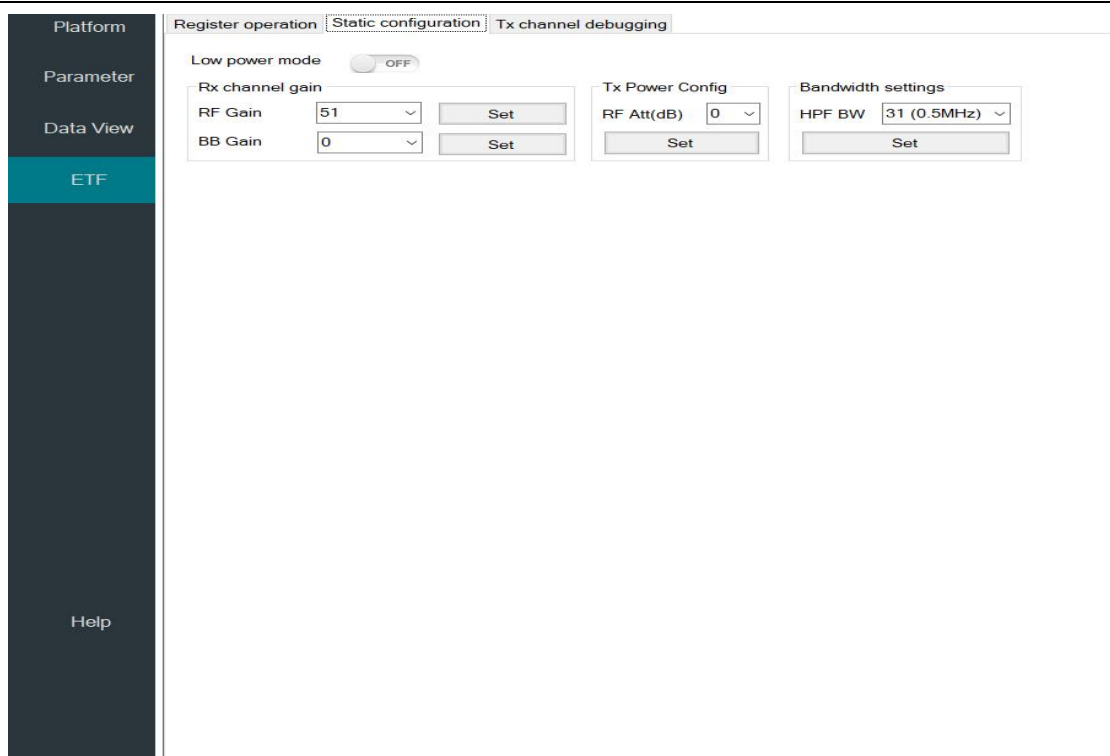


Figure 2-49 Static Configuration Page

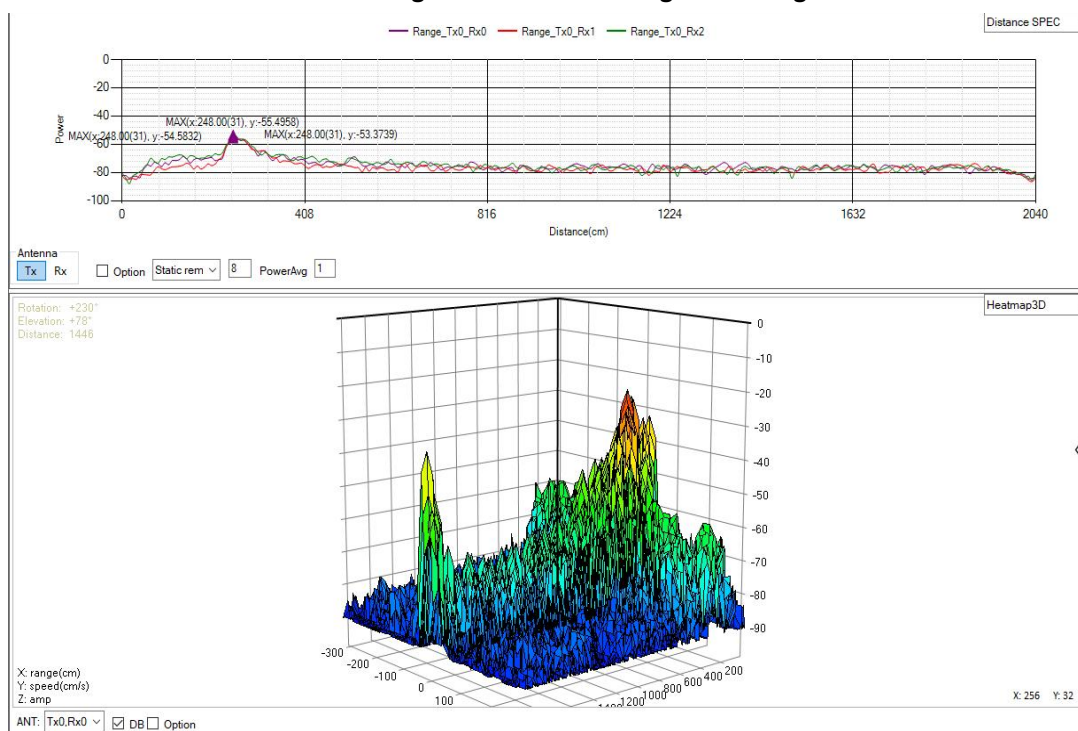


Figure 2-50 Signal Spectrum With Default Static Configuration

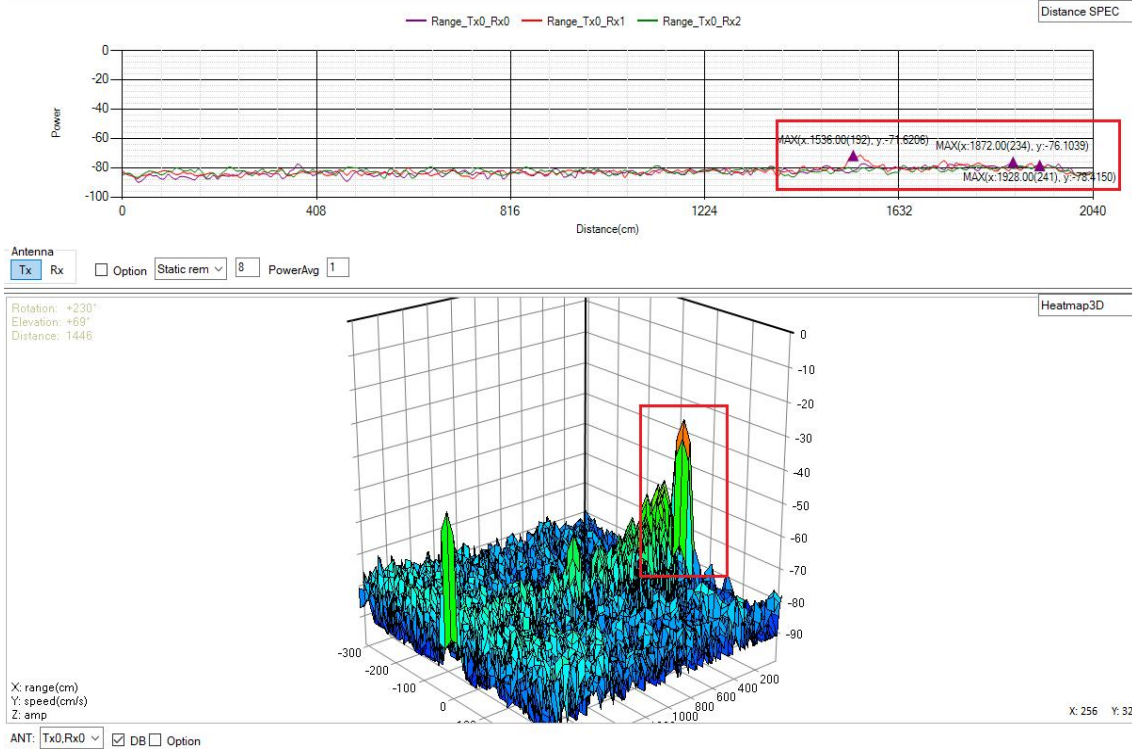


Figure 2-51 Signal Spectrum With Modified Static Configuration

2.7 Script Execution

This section describes the Lua script execution flow.

After connecting to the Device as described in Section 2.1, perform the following steps:

1. Click Select File and select the required Lua script.
2. Click Check Syntax and review the syntax-check result. If no error is reported, as shown in the figure, continue.
3. Click Start Execution. The tool executes the script.
4. Click Stop Execution. The tool stops script execution.

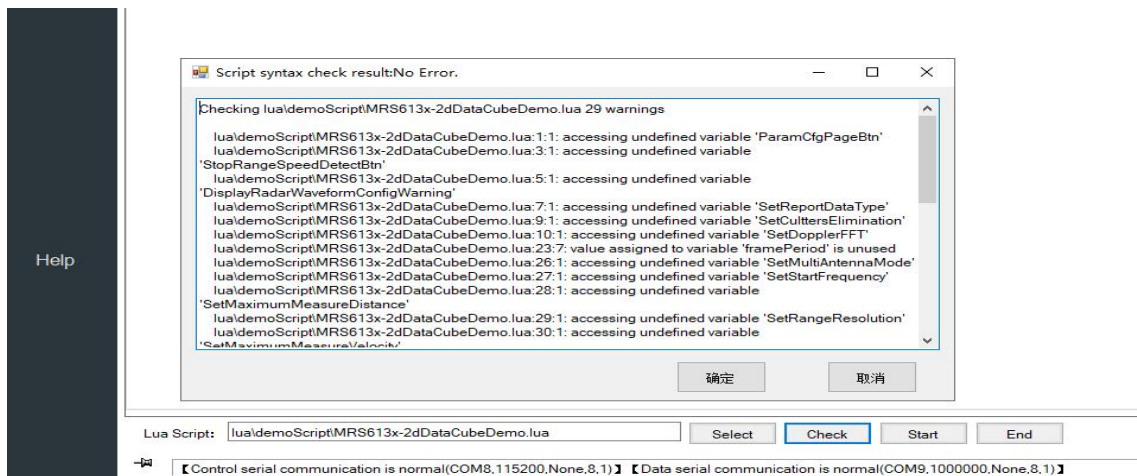


Figure 2-52 Lua Script Execution

2.8 Menu Functions

This section describes the Data Saving, Data Conversion, Log Saving, Screenshot Saving, and BLU939 Power Control menu functions.

2.8.1 Data Save

Data Saving automatically converts captured data to the specified formats while saving it.

After connecting to the Device and starting Data View, perform the following steps:

1. Click Menu to open the menu page.
2. Select one or more save formats. When data is saved, it is also saved automatically in the selected formats.

Note: Multiple formats can be selected. Files in all selected formats are saved automatically in the corresponding folders. Figure 2-54 shows the files saved when only Auto Save FFT Table is selected.

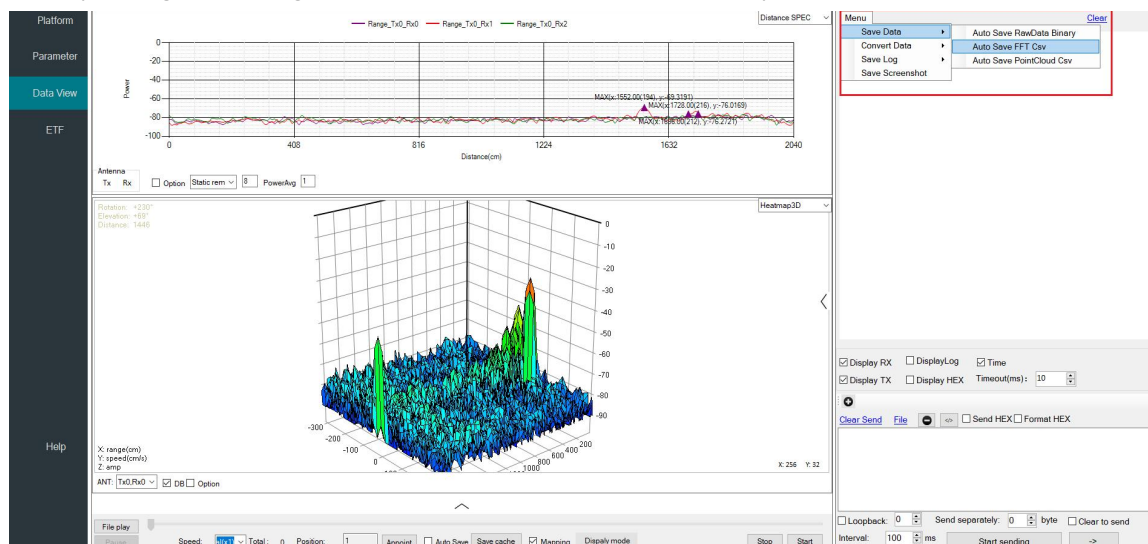


Figure 2-53 Menu: Data Saving

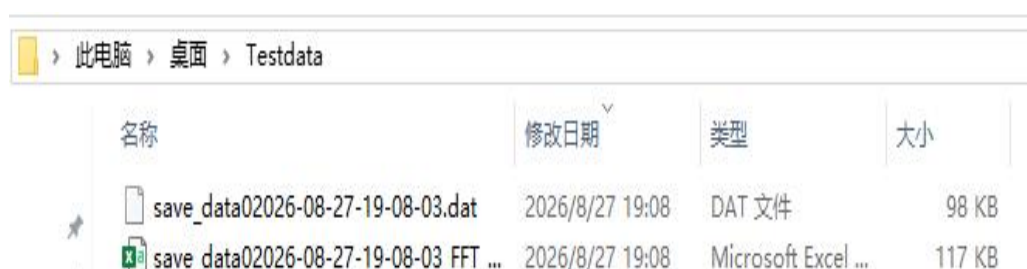


Figure 2-54 File Saving

2.8.2 Data Conversion

Data Conversion converts saved data to another format.

Perform the following steps:

1. Click Menu to open the menu page.
2. Select Data Conversion, select the required output format, and then select the file to convert.

Note: Data Conversion converts the default .dat file to the required format. Only DataCube data can be converted to an FFT table, and only Point Cloud data can be converted to a point-cloud table.

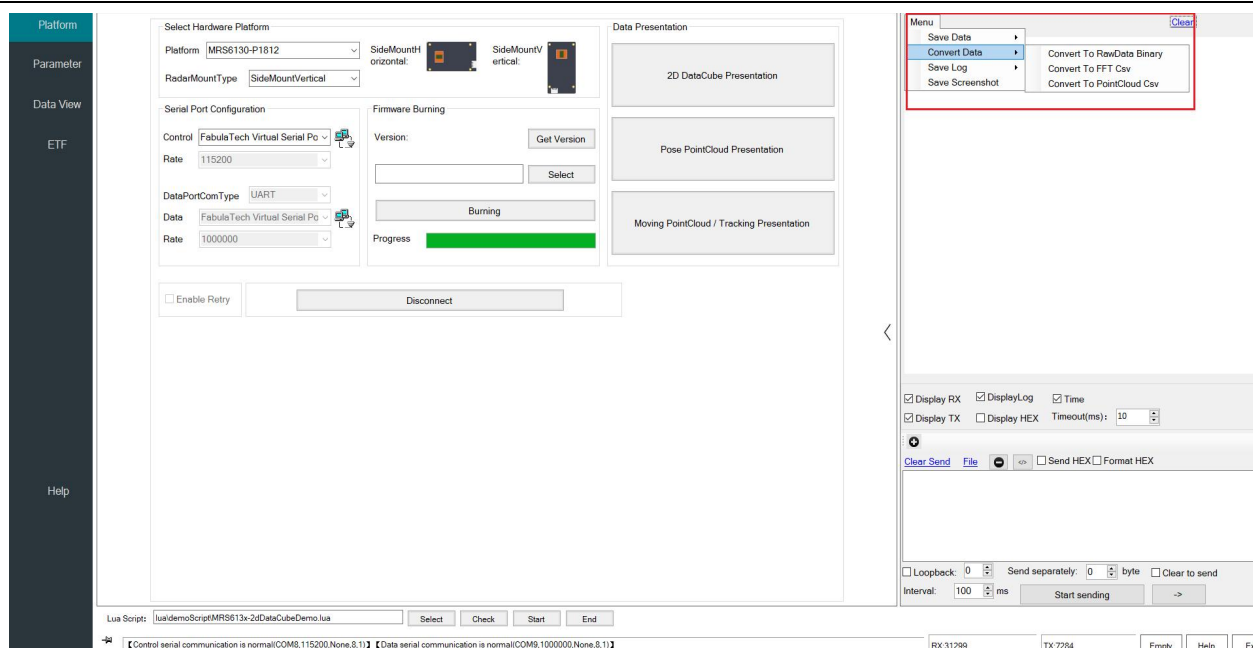


Figure 2-55 Menu: Data Conversion

2.9 Data Formats

2.9.1 FFT Binary File

The FFT binary file extension is .bin. Figure 2-57 shows the file format, and Table 2-4 defines each field.

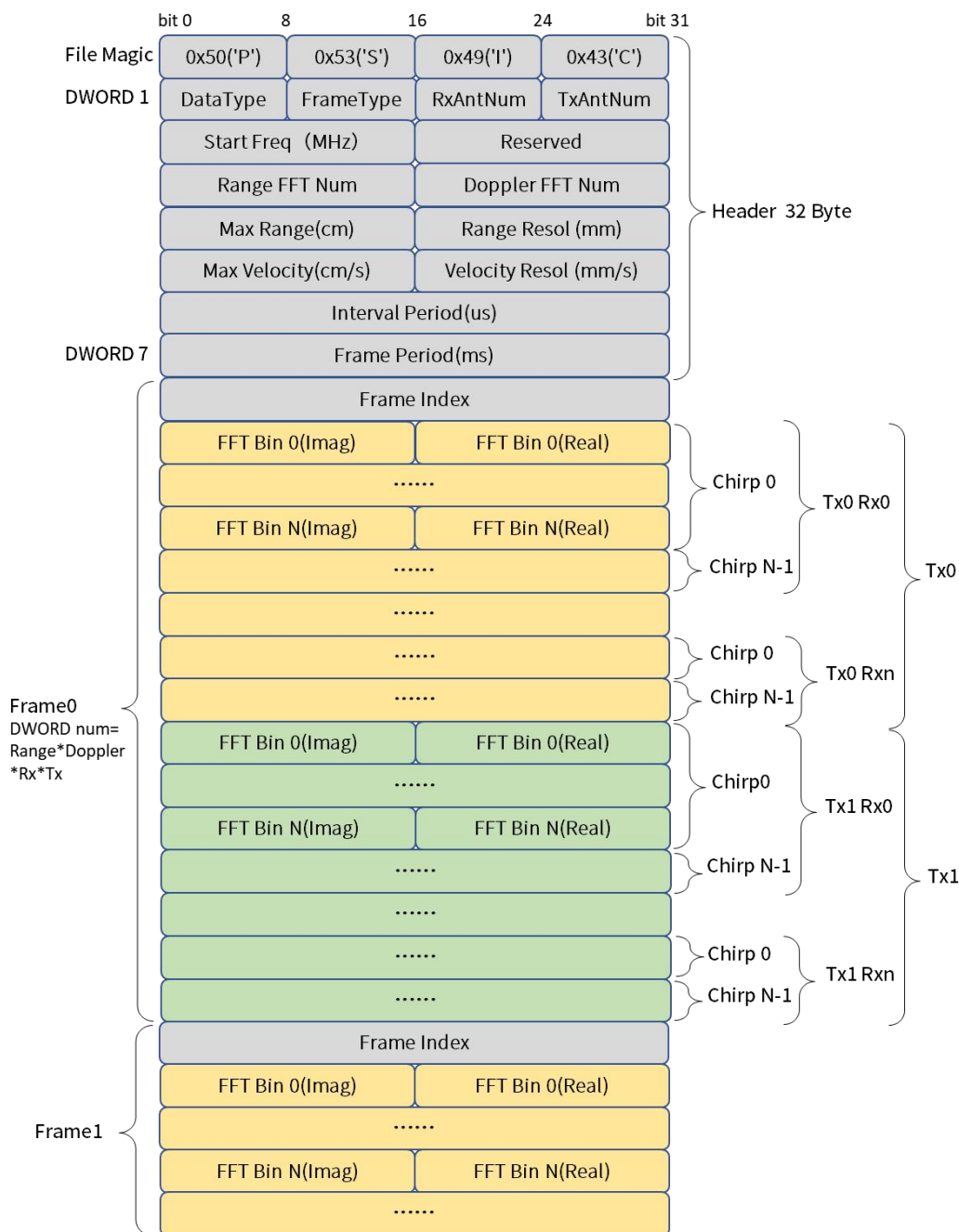


Figure 2-57 FFT Binary Data Format

Table 2-4 FFT Binary Data Format

Frame Parameter	Description
File Magic	File header identifier, fixed as the string "PSIC", with a size of one DWORD.
Data Type	0 indicates FFT data; 4 indicates point-cloud data; and 6 indicates tracking data. Size: 1 byte.
Frame Type	1 indicates 1D FFT; 2 indicates 2D FFT. Size: 1 byte.
RxAntNum	Number of receive channels. Size: 1 byte.
TxAntNum	Number of transmit channels. Size: 1 byte.
StartFreq	Chirp start-frequency configuration, in MHz.
Range FFT Num	Size: 2 bytes.
Doppler FFT Num	Size: 2 bytes.

Frame Parameter	Description
Max Range	Maximum detection range, in cm. Size: 2 bytes.
Range Resol	Range resolution, in mm. Size: 2 bytes.
Max Velocity	Maximum detectable velocity, in cm/s. Size: 2 bytes.
Velocity Resol	Velocity resolution, in mm/s. Size: 2 bytes.
Interval Period(us)	Interval period, in μ s.
Frame Period(ms)	Frame period, in ms.
FrameIndex	Frame index. Size: 4 bytes.
FFT Bin N (Imag)	Imaginary part of FFT data point N: a 16-bit signed integer in the lower 16 bits of the DWORD.
FFT Bin N (Real)	Real part of FFT data point N: a 16-bit signed integer in the upper 16 bits of the DWORD.

2.9.2 FFT CSV File

The FFT CSV file extension is .csv. Figure 2-58 shows the format. FrameIndex is the radar frame sequence number, IntervalIndex is the Interval sequence number for each TX/RX channel, and bin0 through bin255 contain the FFT data for the corresponding Interval. The first row contains FFT real data and the second row contains FFT imag data. Within a frame, IntervalIndex 0 is the first Interval of a TX/RX channel, such as RX0. The receive-channel order is RX0, RX1, and RX2.

FrameIndex	ChirpIndex	bin0	bin1	bin2	bin3		bin255	备注		
0	0	-1	0	1	2		-1	Real	Tx0 Rx0	Frame0
0	0	0	1	2	3		0	Imag		
.....										
0	N-1									
0	N-1									
.....										
0	0									
0	0									
.....										
0	N-1									
0	N-1									
1	0	-1	0	1	2		-1			
1	0	-1	0	1	2		-1			
.....										

Figure 2-58 FFT CSV Data Format

2.9.3 Point-Cloud Binary File

The point-cloud binary file extension is .bin. Figure 2-59 shows the format, and Table 2-5 defines each field. The file header is the same as that of the FFT binary file.

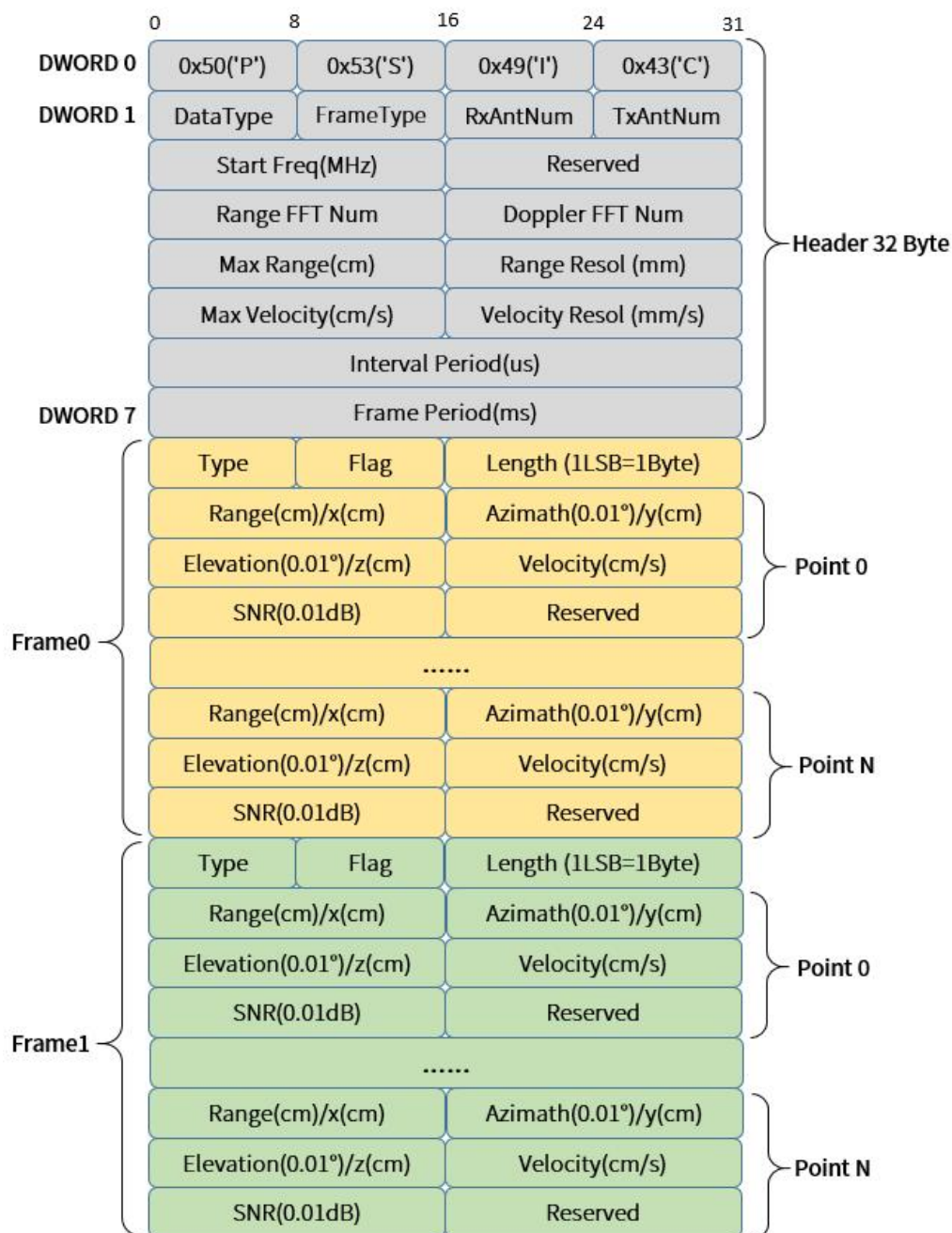


Figure 2-59 Point-Cloud Binary Data Format

Table 2-5 Point-Cloud Binary Data Format

Frame Parameter	Description
File Magic	File header identifier, fixed as the string "PSIC", with a size of one DWORD.
Data Type	0 indicates FFT data; 4 indicates point-cloud data; and 6 indicates tracking data. Type: 8-bit unsigned integer.
Frame Type	1 indicates 1D FFT; 2 indicates 2D FFT. Type: 8-bit unsigned integer.
RxAntNum	Number of receive channels, 8-bit unsigned integer.
TxAntNum	Number of transmit channels, 8-bit unsigned integer.
Start Freq	Chirp start-frequency configuration, in MHz. Type: 16-bit unsigned integer.
Range FFT Num	Range FFT length. Type: 16-bit unsigned integer.
Doppler FFT Num	Doppler FFT length. Type: 16-bit unsigned integer.

Frame Parameter	Description
Max Range	Maximum detection range, in cm. Type: 16-bit unsigned integer.
Range Resol	Range resolution, in mm. Type: 16-bit unsigned integer.
Max Velocity	Maximum detectable velocity, in cm/s. Type: 16-bit unsigned integer.
Velocity Resol	Velocity resolution, in mm/s. Type: 16-bit unsigned integer.
Type	Data type. Type: 8-bit unsigned integer. Fixed at 4 for point-cloud data.
Flag	Data flags. Type: 8-bit unsigned integer. bit0 = 0 indicates polar coordinates; bit0 = 1 indicates Cartesian coordinates.
Length	Data length, in bytes. Type: 16-bit unsigned integer.
Range(cm)/X(cm)	For polar coordinates, this field indicates Range, in cm. Type: 16-bit unsigned integer; For Cartesian coordinates, this field indicates the X coordinate, in cm. Type: 16-bit signed integer.
Azimuth(0.01°)/Y(cm)	For polar coordinates, this field indicates Azimuth, in units of 0.01 degrees. Type: 16-bit signed integer; For Cartesian coordinates, this field indicates the Y coordinate, in cm. Type: 16-bit unsigned integer.
Elevation(0.01°)/Z(cm)	For polar coordinates, this field indicates Elevation, in units of 0.01 degrees. Type: 16-bit signed integer; For Cartesian coordinates, this field indicates the Z coordinate, in cm. Type: 16-bit signed integer.
SNR	Point-cloud SNR, in units of 0.01 dB. Type: 16-bit signed integer.
Reserved	Reserved. Type: 16-bit unsigned integer.

2.9.4 Point-Cloud CSV File

The point-cloud CSV file uses the .csv extension. The file formats are shown in Figure 2-60 and Figure 2-61. FrameIndex is the radar frame sequence number, and PointIndex is the point index within a frame. For polar coordinates, the point-cloud fields are Range, Azimuth, and Elevation, with units of m and degrees. For Cartesian coordinates, the fields are X, Y, and Z, in m. Velocity is in m/s, and SNR is in dB.

Each frame contains 0 to 128 points. The PointIndex of the first point is 0. If a frame contains no points, its frame sequence number does not appear in the file.

FrameIndex	PointIndex	Range	Azimuth	Elevation	Velocity	SNR
0	0	0.76	15.4	-4.4	0.5	10.6
0	1	0.48	-14.8	-36.2	0.2	10.6
1	0	0.52	-11	-33.4	0.2	10.6
2	0	0.56	-18	-20.4	0.2	10.6
2	1	0.6	37.8	49	0.2	10.6
2	2	0.6	33	-49.4	0.3	10.6
2	3	0.64	-13.2	-3.6	0.2	10.6
.....						

Figure 2-60 Polar-Coordinate Point-Cloud CSV Format

FrameIndex	PointIndex	X	Y	Z	Velocity	SNR
0	0	-0.08	0.37	-0.21	0.2	21.4
0	1	-0.08	0.34	-0.25	0.3	21.4
0	2	-0.14	0.43	-0.15	0.2	21.4
0	3	-0.17	0.43	-0.11	0.3	21.4
0	4	-0.08	0.44	-0.25	0.2	21.4
0	5	-0.2	0.45	-0.13	0.3	21.4
1	0	-0.33	0.44	0.04	0.3	19.2
1	1	-0.37	0.43	0.16	0.2	19.2
1	2	-0.34	0.48	0.02	0.5	19.2
1	3	-0.36	0.47	0.05	0.3	19.2
1	4	-0.39	0.48	0.15	0.2	19.2
1	5	-0.37	0.52	0	0.5	19.2
.....						

Figure 2-61 Cartesian-Coordinate Point-Cloud CSV Format

2.9.5 Target CSV File

The target CSV file uses the .csv extension. The file format is shown in Figure 2-62. FrameIndex is the radar frame sequence number, track_uid is the target ID, track_state is the tracking state, and points_num is the number of points forming the target. Target-position fields x, y, and z are in m, and target-velocity fields vx, vy, and vz are in m/s. If a frame contains no target information, its frame sequence number does not appear in the file.

frameIndex	track_uid	track_state	points_num	x	y	z	vx	vy	vz
18	0	1	4	2.62	8.06	0	0	0	0
19	0	1	2	2.47	7.95	0	-0.01	0.04	0
20	0	1	0	2.46	7.96	0	-0.01	0.04	0
21	0	1	4	2.48	7.92	0	-0.01	-0.02	0
22	0	1	2	2.4	7.87	0	-0.09	-0.1	0
23	0	1	2	2.37	7.91	0	-0.11	-0.07	0
24	0	1	5	2.28	7.82	0	-0.16	-0.14	0
25	0	1	4	2.41	7.73	0	0	-0.16	0
26	0	1	0	2.41	7.69	0	0	-0.16	0
27	0	1	13	2.34	7.7	0	-0.04	-0.06	0
27	1	1	8	2.16	5.42	0	0	0	0
28	0	1	1	2.27	7.88	0	-0.1	0.1	0
28	1	1	10	2.19	5.35	0	0	-0.04	0
29	0	1	2	2.19	8	0	-0.17	0.09	0
29	1	1	2	2.23	5.27	0	0.01	-0.09	0
30	0	1	0	2.16	8.02	0	-0.17	0.09	0
30	1	1	6	2.29	5.2	0	0.08	-0.08	0
31	1	1	13	2.36	5.15	0	0.16	-0.05	0
32	2	1	8	-1	1.49	0	0	0	0
.....									

Figure 2-62 Target CSV Data Format

3 Q&A

3.1 Why RadarAnalysisTool Cannot Run

Table 3-1 RadarAnalysisTool Cannot Run

Possible Cause	Corrective Action
Microsoft .NET Framework 4.7.2 is not installed on Windows.	Install Microsoft.NET Framework 4.7.2 correctly.

3.2 Why the Device Serial Port Cannot Be Detected

Table 3-2 Device Serial Port Cannot Be Detected

Possible Cause	Corrective Action
The serial-port driver for the Device is not installed on the current Windows system.	Install the corresponding serial-port driver as prompted by Windows.

3.3 Why the Device Cannot Be Connected

Table 3-3 Device Cannot Be Connected

Possible Cause	Corrective Action
The correct Device serial port is not selected.	Check whether the selected Device serial-port number is correct. Try another port number if necessary.
The correct serial-port baud rate is not selected.	Select the correct baud rate and reconnect.
Poor Device connection.	Check the Device connection and ensure that the Device is operating normally.
Firmware Burn Has Not Been Completed	Read the Device version. If the version cannot be read, firmware burn may not have been performed or the Device may be operating abnormally.

3.4 Why the Data View Updates Too Slowly

Table 3-4 View Updates Too Slowly

Possible Cause	Corrective Action
The serial-port baud rate is too low.	The data port supports a maximum baud rate of 2 Mbaud. A higher baud rate can be selected, but not all Hosts or USB ports support 2 Mbaud. Unless specifically required, use 1 Mbaud.
The Frame Period is configured too large.	Modify Frame Period under Parameter Configuration and confirm the effective Frame Period.
Insufficient PC CPU or memory performance.	Use Windows Task Manager to check whether CPU or memory usage is too high. Close some tasks or use a higher-performance PC.
The number of Doppler points (Intervals) is too large.	Disable the heatmap or reduce the number of Doppler points (Intervals).

Possible Cause	Corrective Action
The Host does not support a baud rate of 2 Mbaud.	Some USB ports do not support 2 Mbaud. If support is unknown, use 1 Mbaud.

3.5 Why RadarAnalysisTool Does Not Save Exported Data to a File

Table 3-5 File Saving Failure

Possible Cause	Corrective Action
Auto Save is not selected under File Playback.	Select the corresponding option.
The corresponding option under Menu > Data Saving is not selected.	Select the corresponding option.
Stop has not been clicked.	Transmission does not stop automatically after all radar frames are sent. Click Stop manually to save the file.
The frame-data type selected for conversion does not match the exported data type	A DataCube frame-data file corresponds to FFT-data export, and a point-cloud frame-data file corresponds to point-cloud data export.

3.6 Why the Radar Point-Cloud View Stops Unexpectedly

Table 3-6 Point-Cloud View Stops

Possible Cause	Corrective Action
The configured number of frames has been transmitted.	Restart transmission.
The Frame Period is too short.	Increase the Frame Period as appropriate and restart transmission.
The module is operating abnormally.	Try restarting the module

3.7 Why Does RadarAnalysisTool Report That the Configured Range Is Too Small?

Do not configure the maximum radar detection range below 5 m or set it exactly equal to the desired detection range. Ensure that the following constraints are satisfied:

Maximum detection range * 0.8 > desired detection range; maximum detection range > 5 m.

Even if the desired maximum detection range is short, configure a larger maximum detection range and remove ranges outside the desired range during subsequent algorithm processing.

3.8 How Is the Minimum Frame Period Configured in RadarAnalysisTool Determined?

Frame Period consists of the transmission duration and estimated reporting time. If the configured Frame Period is too short, RadarAnalysisTool automatically corrects it to the minimum Frame Period. The effective Frame Period must also be an integer multiple of the Interval **Period**.

Appendix A: Radar Shell Commands

Radar Shell Commands is a radar parameter-configuration tool supported by the Device. It configures radar parameters and operates the radar without RadarAnalysisTool, helping evaluate radar performance and analyze solutions and algorithms. The command interface is shown in Figure A-1.

```
uart:~$ mmwc
mmwc - MMW Controller shell commands
Subcommands:
open      : [<xo/msi> [xo/none]]
close     : No param
version   : No param
start     : No param
stop      : No param
state     : No param, Get power state
ps        : <enable>
mode      : <txrx_mode> <FFT mode>
freq      : <start_MHz> [max_MHz]
range     : <max_mm> <resolution_mm>
veloc     : <max_mm/s> <resolution_mm/s>
frame     : <period_ms> <frame_num>
intv      : <period_us> <intv_num>
chirp     : No param
fft_len   : No param
coord     : [polar/cart]
data      : <diabie/fft/3dpoints>
clutter_rm : <enable> [fac]
cfar      : <static_rm> <snr(1~255)>
cube      : [read_num] [offset]
report    : <disable/cube/points> [frame_num]
uart      : <on/off>
baudrate  : <baudrate>
noise     : [est]
tone      : <start/stop> [txid] [freq_MHz]
rx_gain   : <rf/dig> <ant_id> <gain>
calib     : <chirp/load/save>
dbg       : [<type> [tag]]
<Fail 1> Command Failed
uart:~$
```

Figure A-1 Radar Shell Interface

1、Common Startup Procedure

Table A-1 Common Startup Procedure Example

```
mmwc open // Automatically executed at startup in the PointCloud Demo; can be omitted
mmwc mode 1 2 // Configure the antenna mode as 1T3R and the FFT mode as 2D FFT
mmwc range 10000 40 // Maximum range: 10 m; range resolution: 4 cm
mmwc veloc 3000 200 // Maximum velocity: 3 m/s; velocity resolution: 0.2 m/s
mmwc data 3dpoints 4 // Report 3D point-cloud data and print four points to the console
mmwc frame 200 10000 // Frame Period: 200 ms; transmit 10000 frames
mmwc start
```

2、Open the Radar

Usage: mmwc open

Description: Initialize and open the radar controller before any other configuration. Parameters are optional.

3、Close the Radar

Usage: mmwc close

Description: After closing, execute open before configuring or controlling the radar again.

4、Get Firmware Version

Usage: mmwc version

Description: The Shell prints the radar firmware version.

5、Start Detection

Usage: mmwc start

Description: Configure radar parameters before executing Start, except for the power-saving configuration. If no parameters are configured, the defaults are used.

6、Stop Detection

Usage: mmwc stop

Description: To update radar parameters, execute Stop before reconfiguration.

7、Get State

Usage: mmwc state

Description: Four states are available: power down, active, sleep, and running.

8、Power-Saving Configuration

Usage: mmwc ps <enable>

Description: enable = 0 disables system power saving; enable = 1 enables system power saving, allowing the system to enter sleep mode between frames.

Note: The point-cloud demo firmware does not support power saving by default. To enable it, refer to the low-power development guide and upgrade the firmware.

9、Operating Mode

Usage: mmwc mode <TxRx mode> <FFT mode>. If no parameters are specified, the current configuration is returned.

- 1) TxRx mode is the antenna mode. Type: 8-bit unsigned integer. 0 indicates 1T1R; 1 indicates 1T3R.
- 2) FFT mode is the FFT mode. Type: 8-bit unsigned integer. 1 indicates 1D FFT; 2 indicates 2D FFT.

Example: mmwc mode 0 1 configures 1T1R and 1D FFT mode.

10、Sweep Frequency

Usage: mmwc freq <start freq> [max freq]. If no parameters are specified, the current configuration is returned.

- 1) start freq is the start frequency of the sweep, in MHz. Type: 32-bit integer.
- 2) max freq is the optional maximum frequency of the sweep, in MHz. Type: 32-bit integer.

11、Detection Range

Usage: mmwc range <max> <resolution>. If no parameters are specified, the current configuration is returned.

- 1) max is the maximum range, in mm. Type: 32-bit unsigned integer.
- 2) resolution is the range resolution, in mm. Type: 32-bit unsigned integer.

Note: FFT bin size (range difference between adjacent FFT bins) = maximum range / actual FFT length (aligned to a power of two).

12、Detection Velocity

Usage: mmwc veloc <max> <resolution>. If no parameters are specified, the current configuration is returned.

- 1) max is the maximum velocity, in mm/s. Type: 32-bit unsigned integer.
- 2) resolution is the velocity resolution, in mm/s. Type: 32-bit unsigned integer.

Note: FFT bin size (velocity difference between adjacent FFT bins) = maximum velocity / actual Doppler FFT length (aligned to a power of two).

13、Frame Configuration

Usage: mmwc frame <period> <num>. If no parameters are specified, the current configuration is returned.

1) period is the frame-transmission period, in ms. Type: 32-bit unsigned integer.

2) num is the number of frames. Type: 32-bit unsigned integer.

14、 Interval Configuration

Usage: mmwc intv <period> <num>. If no parameters are specified, the current configuration is returned.

1) period is the Interval Period, in us. Type: 32-bit unsigned integer.

2) num is the number of Intervals. Type: 32-bit unsigned integer.

Note: Configurable only in 1D FFT mode.

15、 Chirp Parameter Display

Usage: mmwc chirp

Description: Returns the current Chirp configuration. This configuration is calculated from the range settings.

16、 FFT Length Parameter Display

Usage: mmwc fft_len

Description: Returns the current range FFT and Doppler FFT lengths. This configuration is calculated from the range and velocity settings.

17、 Coordinate-System Configuration

Usage: mmwc coord <type>. If no parameters are specified, the current configuration is returned.

1) type is the coordinate-system type. Type: string. "polar" indicates polar coordinates; "cart" indicates Cartesian coordinates.

18、 Static-Clutter Filtering Configuration

Usage: mmwc clutter_rm <enable> [fac]. If no parameters are specified, the current configuration is returned.

1) enable controls filtering. Type: Boolean. 1 enables filtering; 0 disables filtering.

2) fac is the static-clutter update coefficient and controls convergence. A larger value produces slower convergence. Type: unsigned integer; range: 0 to 16. If omitted, the previous setting is used. Default: 0.

19、 CFAR Threshold Filtering Configuration

Usage: mmwc cfar <static_rm> <snr>. If no parameters are specified, the current configuration is returned.

1) static_rm specifies whether to remove the zero-Doppler bin and the adjacent +/-1 bins for targets stationary relative to the radar. 1 removes them; 0 retains them.

2) snr is the CFAR SNR threshold. Type: 8-bit unsigned integer; range: 1 to 255; default: 80.

20、 Read DataCube Data

Usage: mmwc cube [read_num] [offset]. If no parameters are specified, the defaults are used.

1) read_num is the number of DataCube points to read. Default: 8.

2) offset is the DataCube starting-point offset. Default: 0.

21、 Data Printing

Usage: mmwc data <print_type> <print num>

Description: Data Printing outputs radar data to the console. It is typically used for debugging without a connected Host and provides a simple data display.

1) print_type is the data-output type. Type: string. "disable" disables data output; "fft" outputs FFT data; "3dpoints" outputs 3D point-cloud data.

2) print num is the number of data items to print. Type: 32-bit unsigned integer.

22、 Data Reporting

Usage: mmwc report <data_type> <frame num>

Description: Data Reporting sends radar data through the SERIAL-A interface. It is generally used by RadarAnalysisTool to display radar data or by the Host for real-time processing.

1) data_type is the report data type. Type: string. "disable" disables data reporting; "cube" reports FFT data; "points" reports point-cloud data.

2) frame num is the number of frames to report. Type: 32-bit unsigned integer.

23、Data UART Switch

Usage: mmwc uart <on/off>

Description: Controls the UART used for data reporting. Enable this UART when data reporting is required.

1) on/off controls the UART. Type: string. "on" enables the data UART; "off" disables it.

24、Data Baud-Rate Setting

Usage: mmwc baudrate <baudrate>. If no parameter is specified, the current configuration is returned.

1) baudrate sets the SERIAL-A baud rate. Type: 32-bit unsigned integer. Maximum supported baud rate: 2000000.

Note: The baud rate must match the Host setting; otherwise, data cannot be reported.

25、Noise Estimation

Usage: mmwc noise [est]. If no parameter is specified, the current estimated noise is returned.

1) est starts noise estimation. Type: string.

26、Tone Transmission

Usage: mmwc tone <start/stop> [txid] [freq_MHz]. If no parameters are specified, the current configuration is returned.

1) start/stop controls tone transmission. Type: string. "start" starts transmission; "stop" stops transmission.

2) txid is the TX channel for tone transmission. Type: unsigned integer. 255 indicates all channels. Default: 255.

3) freq_MHz is the tone frequency, in MHz. Type: unsigned integer. Default: 58000 MHz.

27、RX Gain Configuration

Usage: mmwc rx_gain <rf/dig> <ant_id> <gain>.

1) rf/dig selects the gain type. Type: string. "rf" indicates analog gain; "dig" indicates digital gain.

2) ant_id is the RX channel to configure. Type: unsigned integer. 255 indicates all channels.

3) gain is the configured gain, in dB. Type: signed integer. Note: Gain values are discrete. RF gain supports only 27, 33, 39, 45, and 51; digital gain supports only -18, -12, -6, 0, 6, 12, and 18.

28、Chirp Calibration

Usage: mmwc calib <chirp/load/save>.

1) chirp/load/save selects the calibration operation. Type: string. "chirp" reruns Chirp calibration; "load" loads calibration data from Flash; "save" saves the current calibration data to Flash.

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